

Pluton Formation and the Volcanic–Plutonic Connection in Large Igneous Provinces: Does the Brandberg Granite Intrusion Mark the Extinction of Etendeka Supervolcanoes?

YI SUN¹, ANDREA GALLI¹, DAWID SZYMANOWSKI¹, MARCEL GUILLONG¹, JEREMIA SIMON², ABRAHAM SHIPANDENI² and OLIVIER BACHMANN¹

¹Institute of Geochemistry and Petrology, Department of Earth and Planetary Sciences, ETH Zurich, Clausiusstrasse 25, 8092 Zurich, Switzerland

²Geological Survey of Namibia, Ministry of Energy and Mines, 1 Aviation Road, 2168 Windhoek, Namibia

*Corresponding author. Telephone: +41 44 632 78 23. E-mail: yi.sun@erdw.ethz.ch

The northwestern Namibian section of the Cretaceous Paraná-Etendeka Large Igneous Province (LIP) displays granitic complexes and voluminous, coeval ignimbrite deposits, providing a rare opportunity to study silicic magmatism in an LIP-related setting at two different crustal levels: the upper crust and the surface. Here, using high-precision zircon geochronology, petrology and geochemistry, we focus on the ~2000 km³ Brandberg intrusion and two of the most voluminous ignimbrites of the Northern Etendeka area, the ~8600 km³ Sarusas and ~4000 km³ Khoraseb quartz latites. Brandberg zircon crystals record incremental pluton growth over ~1.4 Myr, beginning with the intrusion of the main granite body at 133.20 ± 0.19 to 132.62 ± 0.29 Ma, followed by the emplacement of quartz monzonites and peripheral dykes at 132.733 ± 0.042 to 132.14 ± 0.18 Ma, and finally the highly fractionated, fluid-enriched and partly pegmatitic peralkaline Amis granite from 131.910 ± 0.039 to 131.838 ± 0.043 Ma. Field observations suggest incremental emplacement of silicic magmas at shallow subvolcanic crust levels following a caldera collapse event, but there is no preserved evidence for chemically evolved volcanic deposits around Brandberg that could be related to a caldera-forming eruption or any other volcanic event during the pluton assembly. In the upper part of the Northern Etendeka volcanic stratigraphy, the two quartz latite ignimbrites Khoraseb (133.319 ± 0.073 Ma) and Sarusas (132.572 ± 0.034 Ma) are coeval to the Brandberg intrusion. These hot, unzoned and crystal-poor ignimbrites are comparable to the main units of the Brandberg pluton in terms of major and trace elements. Hence, the Brandberg pluton cannot be representative of a cumulate left-over of volcanic units such as the Khoraseb and Sarusas, but must have accumulated in the upper crust without much loss to the volcanic record (yielding a very low volcanic/plutonic ratio) during the late stages of the Paraná-Etendeka LIP evolution, as the crust was thermally mature. The estimated fluxes of silicic magmas into the Brandberg magma reservoir are relatively low (~10⁻³ km³/yr), consistent with a waning magmatic source and leading to crustal conditions that were very favourable for magma entrapment in the crust. The large volumes of quartz latites erupted from magma reservoirs that are not exposed as plutonic remnants in the Northern Etendeka today; such crustal reservoirs were likely deeper than the Brandberg pluton, which allowed the accumulation of large amounts of hot, crystal-poor melt and avoided the collapse of large caldera structures.

Key words: CA-ID-TIMS; large igneous provinces; magmatic fluxes; plutonic roots; volcanic–plutonic connection

INTRODUCTION

The Early Cretaceous (~135–132 Ma) Paraná-Etendeka Large Igneous Province (LIP), spanning Brazil and Namibia, is known for producing some of Earth's largest ignimbrite deposits. This province witnessed at least five silicic super-eruptions, each with estimated eruptive volumes between ~4000 and 8600 km³ dense rock equivalent (DRE), including the Guarapuava-Tamarana/Sarusas, Santa Maria/Fria, and PAV B-Caxias do Sul/Springbok systems (Milner *et al.*, 1995; Marsh *et al.*, 2001; Bryan *et al.*, 2010). In Namibia's Etendeka region, these volcanic deposits are complemented by extensive plutonic complexes, such as Brandberg, Erongo, and Messum, that are collectively known as the Damaraland complexes. These intrusions, dominated by granites and syenites with minor gabbros and carbonatites, reach

up to 30 km in diameter (Trumbull *et al.*, 2000; Sun *et al.*, 2024) and provide critical insights into fossil upper-crustal magma reservoirs that may have fed the LIP's silicic volcanism.

The Paraná-Etendeka LIP formed through a combination of lithospheric extension preceding South Atlantic rifting and the influence of the Tristan-Gough mantle plume (Hawkesworth *et al.*, 2000; Marsh *et al.*, 2001; Beccaluva *et al.*, 2020; Sun *et al.*, 2024). Namibia's arid climate and prolonged tectonic stability have exceptionally preserved both the crystal-poor ignimbrites and their intrusive counterparts (Renne *et al.*, 1996; Trumbull *et al.*, 2000), offering a rare opportunity to study volcanic–plutonic connections in a continental LIP.

The origin of high-volume, high-temperature, crystal-poor silicic magmas remains debated (Ewart *et al.*, 1998b, 2004b; Ellis

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et al., 2013, 2014; Wotzlaw et al., 2014). In anorogenic settings like LIPs, assimilation-fractional crystallization (AFC) dominates crustal magma evolution in the genesis of high-temperature rhyolites (Bonin, 2007; Natali et al., 2011; Halder et al., 2021; Moya et al., 2021), and silicic super-eruptions should leave behind substantial plutonic residues (Bachmann & Huber, 2016 and references therein). However, such plutonic roots are typically deeply buried, making the well-exposed Damaraland complexes a unique natural laboratory to investigate these processes.

Previous studies on the Etendeka's volcanic-plutonic linkages have focused on Damaraland intrusions with well-preserved volcanic cover, such as Paresis and Erongo (Pirajno et al., 2000; Wigand et al., 2004), or on locating eruptive vents via geochemical and spatial correlations (Milner & Ewart, 1989; Owen-Smith et al., 2017). The Brandberg complex, one of the Damaraland's largest (~21 × 24 km) and best-preserved granite pluton, has received attention for its intrusive history (Diehl, 1990; Schmitt et al., 2000), but its relationship to Etendeka volcanism remains unclear due to extensive erosion of potential volcanic cover (only small remnants exist as nearly inaccessible roof pendants). Existing geochronology relies on limited $^{40}\text{Ar}/^{39}\text{Ar}$ dating (Schmitt et al., 2000), leaving key questions about its magmatic evolution unresolved.

Here, we present high-precision U–Pb zircon geochronology, Hf isotope data, and bulk-rock geochemistry from the Brandberg complex, alongside the first high-precision zircon ages for two major Northern Etendeka quartz latite units (Sarusas, Khoraseb). These data allow us to (1) reconstruct Brandberg's magmatic lifespan, (2) examine the mechanisms behind large silicic pluton formation in LIP settings, and (3) explore potential genetic ties between upper-crustal (Brandberg) and surface (quartz latites) silicic magmas within the same province.

GEOLOGICAL SETTING

The Brandberg complex in Central to NW Namibia intruded into Late Pan-African Damara granites and metasedimentary rocks, as well as Permian–Triassic volcanic/sedimentary rocks of the Karoo Supergroup (Figs 1A and B and 2A). The complex consists mainly of subalkaline, ferroan and metaluminous A-type granites to quartz monzonites (Diehl, 1990; Schmitt et al., 2000). Peralkaline granites and pegmatites occur within the Amis complex (4 × 2 km in size), a layered peripheral intrusion at the SW margin of the main pluton. Additionally, a rounded plug of quartz monzonite (2.5 × 2.5 km in size) is exposed near the central part of the main pluton (Diehl, 1990). The granitic basement NE of Brandberg is intruded by a swarm of peripheral trachydacite dykes. At the S and W peripheries, the outer boundary of the massif is developed as a ring fault system with a narrow, highly brecciated fracture zone extending up to ~500 m into the surrounding country rocks. This ring fault is interpreted as the remains of a major caldera collapse predating the granitic body, but no volcanic/magmatic products from this early activity remain in the present rock record, except for potential remnants in the roof pendants (Korn & Martin, 1953; Diehl, 1990). Outside of the ring fault, the foothills bordering the pluton consist of Early Cretaceous Etendeka volcanic successions, which cap the strongly faulted, Permian–Triassic Karoo sedimentary country rocks over an unconformity (Fig. 2B). The volcanic-sedimentary strata dip progressively towards the granite body with up to ~90° dips (Fig. 2B; Schmitt et al., 2000).

From his geologic mapping of the Brandberg complex, Diehl (1990) proposed eight successive magmatic events based on field relationships: (1) intrusion of trachydacite dykes; (2) formation of a master caldera fault surrounding the main granite body;

(3)–(6) intrusion of four successive main granite units including the edenite–augite granite, the edenite–biotite granite, and two central biotite-bearing granites; (7) intrusion of the central Naib quartz monzonite plug; and finally (8) intrusion of the peralkaline granite and pegmatite of the Amis complex. Due to the geochemical and petrographic similarity of the main granite units, a simplified model comprising four intrusive units was developed by Schmitt et al. (2000): (1) early quartz monzonite intrusion; (2) emplacement of the biotite–hornblende granite body; (3) emplacement of the peralkaline granite and pegmatite; (4) intrusion of the peripheral trachydacite dykes. Since we recognized the units mapped out by Diehl (1990) along our sampling profiles, we adhere to the older unit names in this paper. $^{40}\text{Ar}/^{39}\text{Ar}$ dating of biotite in the Brandberg main granite and arfvedsonite/astrophyllite in the Amis peralkaline granite gave overlapping ages between 133 and 130 Ma (Schmitt et al., 2000).

The Etendeka volcanic deposits in central to NW Namibia (Fig. 1A) extend over a surface area of ~12 000 km² from the Atlantic coast up to 120 km inland, reaching a maximum thickness of nearly 900 m towards the eastern margin (Milner et al., 1992; Ewart et al., 1998a). To the north, the outcrop area extends to Cape Fria and into Angola (Marsh & Swart, 2018). The bimodal Etendeka volcanic province comprises mostly mafic to intermediate rocks (48–58 wt % SiO₂) and associated silicic volcanic rocks (65–72 wt % SiO₂) (Fig. 3; Marsh et al., 2001). The latter correspond to dacite or rhyolite based on the total alkali versus silica classification (Fig. 3A), but they are commonly referred to as 'latites' and 'quartz latites' due to their unusually high K, Fe and low Al bulk compositions (Marsh et al., 2001). As the rock names 'latite' and 'quartz latite' for the Etendeka silicic rocks are well-established in the literature, we follow the nomenclature throughout the present paper. The Etendeka quartz latites are interpreted as high-grade ignimbrites that erupted at unusually high magmatic temperatures for silica-rich melts (ca. 875–980 °C) based on pyroxene geothermometry performed on augite, pigeonite and orthopyroxene (Ewart et al., 2004b) and indicated by their highly welded, rheomorphic textures (Milner et al., 1992).

The Etendeka province shows a pronounced north–south division based on bulk-rock Ti and trace element contents, which can be observed in the both the silicic and mafic units: the high-Ti series is restricted to the northwestern part of the province, whereas the low-Ti series is predominantly present in the southeastern part (Ewart et al., 1998a, 2004a, 2004b; Marsh et al., 2001). In the Southern Etendeka region, silicic volcanic rocks (low-Ti series) are present in the Goboboseb Mountains located around the Messum intrusive complex, and in two domains to the north and south of the Huab River (Jerram et al., 1999). In these outcrops, up-to-300-m-thick Etendeka basalt flows are intercalated with or capped by intermediate to silicic volcanic units including the Goboboseb, Springbok and Grootberg quartz latites (Ewart et al., 1998a; Jerram et al., 1999). The Northern Etendeka region is located ~300 km NW of the Brandberg complex (Fig. 1A). It consists of two coastal domains, both containing silicic volcanic rocks (mostly high-Ti series), and an inland domain dominated by the Khumib block, in which only mafic lavas are exposed (Marsh et al., 2001; Ewart et al., 2004a). In the coastal domains, latites and quartz latites crop out within a 400–500-m thick succession interbedded with minor mafic lavas and are overlying the high-Ti mafic lava flows of the Khumib Formation (Ewart et al., 2004b). The volcanic sequence (~1 km thick) is underlain by Mesozoic sandstones and dips dominantly towards the east and south (Ewart et al., 2004b). The volcanic rocks show locally intense faulting by mainly NW–SE-orientated normal faults (Ewart et al., 2004b).

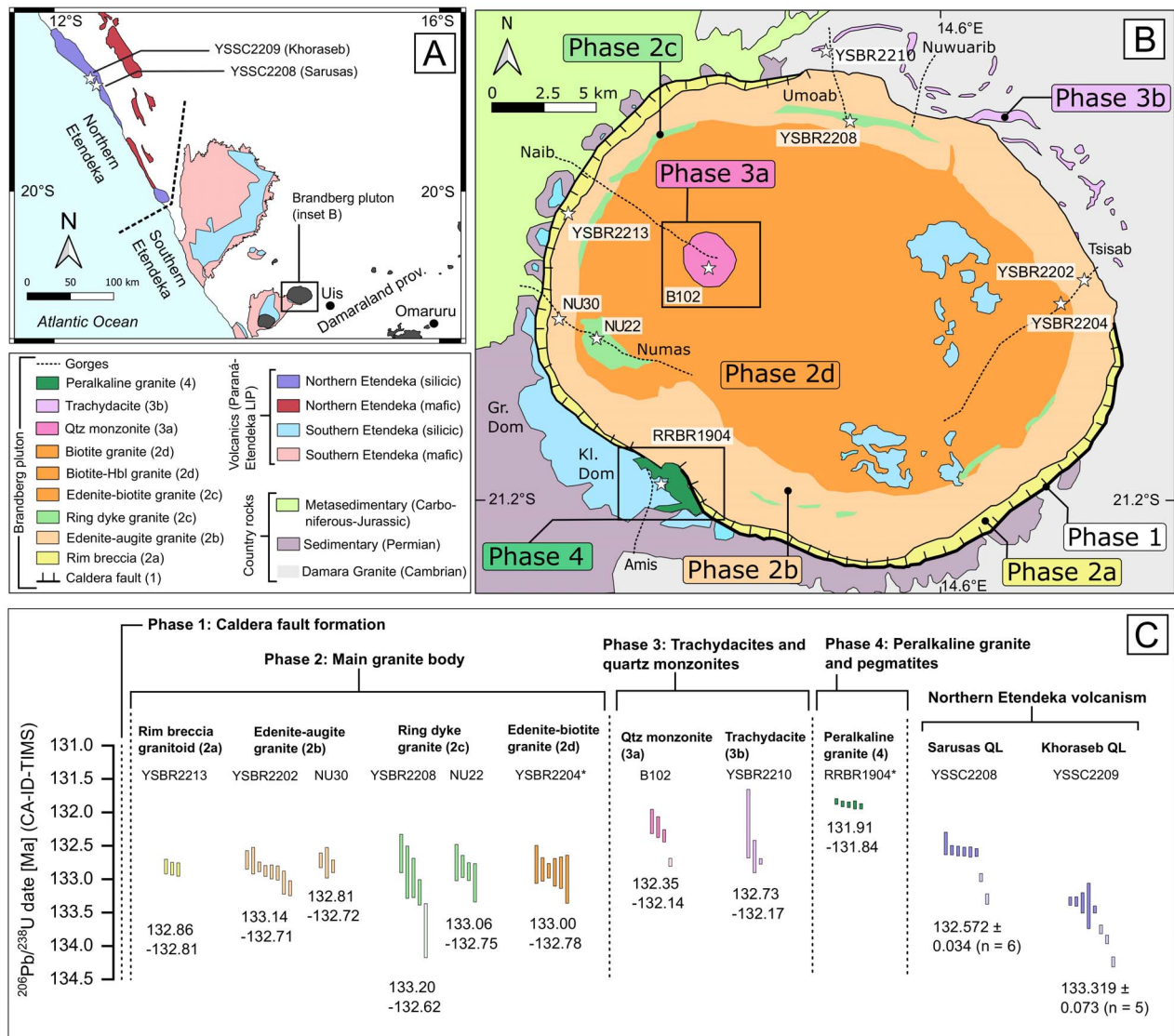


Fig. 1. (A) Map of the Etendeka LIP showing the volcanic outcrops and sampling sites. The dashed line marks the boundary between the Northern and Southern Etendeka sub-provinces (Ewart *et al.*, 2004a). Outcrops of the Early Cretaceous Damaraland intrusions are in dark grey. The box marks the location of the Brandberg pluton. (B) Geological map of the Brandberg pluton with the phases of its evolution (phases 1–4; modified after Diehl, 1990) and sampling locations of all rocks used for high-precision dating. Samples NU30 and NU22 are described in Schmitt *et al.* (2000). (C) Zircon U–Pb dates determined by CA-ID-TIMS for Brandberg plutonic rocks and Northern Etendeka quartz latites with uncertainties plotted as 95% confidence intervals. Bayesian eruption ages are stated for all volcanic samples. Samples marked with stars were previously described in Sun *et al.* (2024), and zircon dates plotted in unsaturated colours indicate grains interpreted as antecrysts.

Towards the eastern boundary, the volcanic rocks are downfaulted against the Damara basement along a major detachment surface (Ewart *et al.*, 2004a).

The Northern Etendeka latites and quartz latites form three suites: the small-scale Nadas latite is the geochemically most primitive and stratigraphically oldest, followed by the slightly more voluminous Sechomib-Hoarusib-Ventura suite (Ewart *et al.*, 2004b). The voluminous ignimbrites of the Khoraseb-Sarasus suite (4000–8600 km³), together with the thick low-Ti Fria quartz latite that lies stratigraphically between the Khoraseb and Sarasus units, occur near the top of the stratigraphic column and therefore represent the last major units to erupt (Ewart *et al.*, 2004b). They are overlain only by the very small-volume Naude quartz latite, and, in the southernmost part of the province, by late-erupted layers of the Terrace quartz latite (Ewart *et al.*, 2004b). The Sarasus quartz latite (Fig. 2E) reaches up to 120 m thickness

and can be traced over 200 km along the Namibian coast (Ewart *et al.*, 2004b). This unit is an important stratigraphic marker for the Northern Etendeka region and has been geochemically correlated with the Guarapuava trachydacite in Brazil, specifically the Tamarana sub-group (Marsh *et al.*, 2001; Ewart *et al.*, 2004b).

MATERIAL AND METHODS

Sampling strategy

Thirty-three samples of Brandberg granite, nearby volcanic rocks, and Northern Etendeka quartz latites were collected in Namibia during a field campaign in 2022. Sampling of the Brandberg aimed at covering all major petrographic units, and zircon separated from the Brandberg quartz monzonite, edenite-augite granite, edenite-biotite granite, ring-dyke granite, and the trachydacite

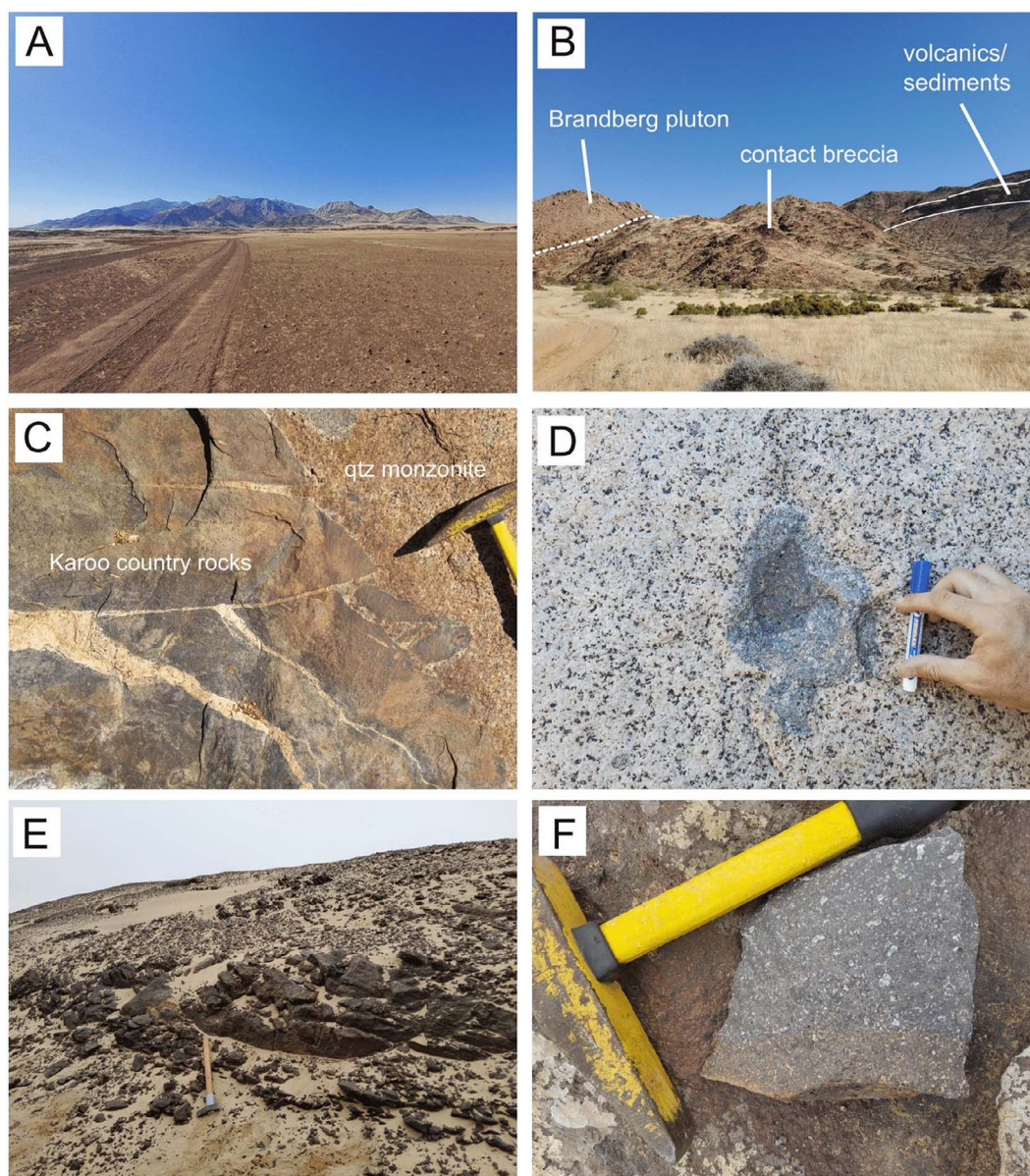


Fig. 2. Field photographs showing (A) the Brandberg granite pluton, viewed from the SW towards Amis gorge; (B) the outer rim breccia zone at western Brandberg in the entrance to Numas valley, showing conformably bedded Etendeka volcanic-sedimentary sequences overlying strongly folded metasedimentary rocks, and dipping towards the pluton; (C) quartz monzonite intruding metasedimentary country rocks and showing signs of magmatic stoping in the breccia zone near Klein-Numas; (D) edenite-augite granite containing a centimetric-sized mafic enclave rimmed by microgranite at Tsisab gorge, ~650 m NE of White Lady rock painting site; (E) outcrop of the rheomorphic Sarusas quartz latite ignimbrite; (F) hand specimen of a moderately crystal-rich Khoraseb quartz latite with large plagioclase phenocrysts and reddish-brown alteration rind.

(Fig. 1B) were investigated. As the central parts and higher altitudes could not be sampled due to the terrain difficulty, black sand containing detrital zircon was collected near the rim of the complex from two dry rivers. These dry rivers originate from the central part of the complex and should therefore carry zircon grains from all Brandberg units exposed at the surface, including the central biotite-hornblende granite not covered by rock sampling. The zircon age distributions for these detrital samples were used as reference to check for the representativeness of our zircon pickings from the rock samples. In addition, three granite samples from the southern part of the Brandberg complex were obtained from the collections of ETH Zurich, and additional three samples from the western and central part were provided from collections of GFZ Potsdam. Zircon and bulk-rock data for the edenite–

biotite granite and the Amis peralkaline granite were previously published in Sun *et al.* (2024); data from these units are included here to provide a complete image of Brandberg magmatism. The volcanic country rocks from the foothills around the Brandberg complex were sampled to identify coeval deposits that could have originated from the complex itself. However, none of the collected samples yielded zircon.

In the Etendeka silicic volcanic rocks, zircon is not ubiquitous. Preliminary heavy mineral separation from three Southern Etendeka quartz latite units (Goboboseb, Spingbok, Grootberg) with Zr contents of 260–350 $\mu\text{g/g}$, the highest we observed in sample material from collections of ETH Zurich, did not yield any zircon grains. Our sampling therefore focused entirely on the Northern Etendeka, where silicic volcanic rocks of the

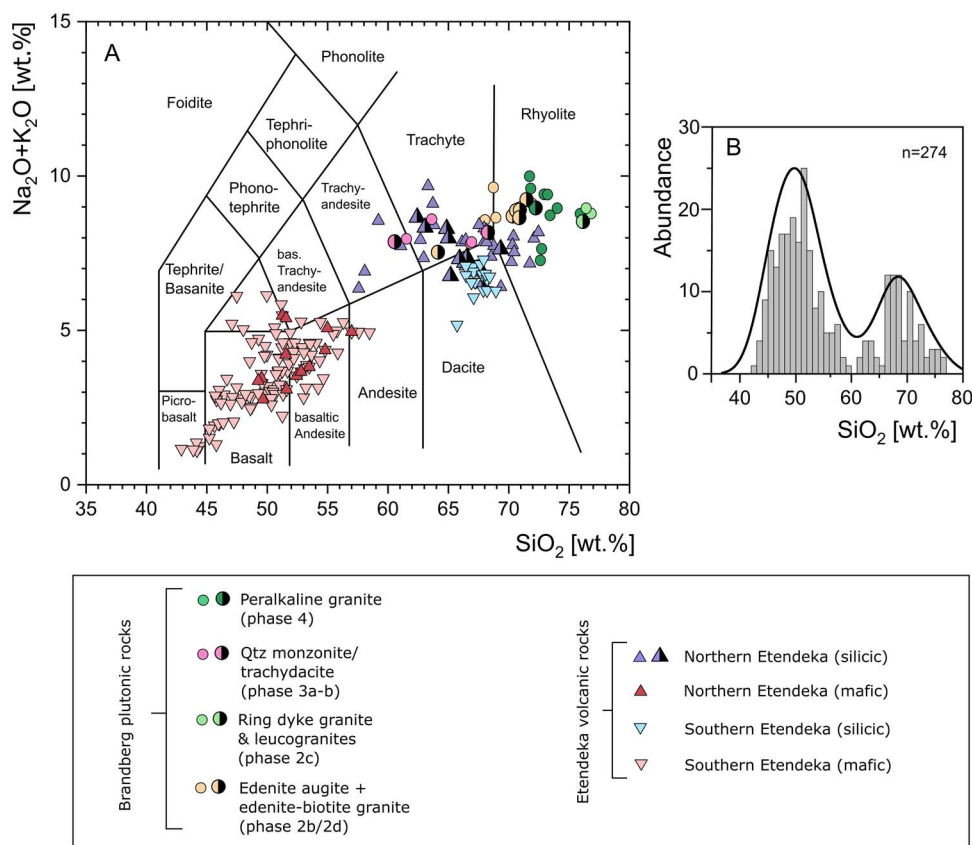


Fig. 3. (A) Total alkalis versus silica diagram for Brandberg and Etendeka igneous rocks. Half-filled symbols are original data from this study, and fully filled symbols are literature data compiled from Bowden *et al.* (1990), Ewart *et al.* (1998a, 1998b) (average values), Ewart *et al.* (2004a, 2004b) (values), Marsh *et al.* (2001), Schmitt *et al.* (2000), and Zhou *et al.* (2020). (B) Histogram of silica contents in Etendeka volcanic rocks and Damaraland plutonic rocks showing the Daly gap for intermediate magma compositions in the Etendeka LIP.

high-Ti suite have higher Zr concentrations (Ewart *et al.*, 2004b) and are, therefore, more likely to have reached zircon saturation. Eleven samples covering seven different quartz latite units (Terrace, Ventura, Khoraseb, Elliot, Fria, Sarusas and Naude) were collected from the coastal outcrops. Of these samples, zircon was found only in the Sarusas and Khoraseb quartz latites. Sampling locations and descriptions are supplied in [Supplementary Table S1](#).

Phase mapping of thin sections

Polished thin sections of 50 μm thickness were prepared from the rock samples and coated with a $\sim 20\text{-nm}$ conductive carbon coating at the Institute of Geochemistry and Petrology, ETH Zurich. Quantitative phase mapping of whole thin sections was carried out using a JEOL JXA-8230 Electron Probe Microanalyzer (EPMA) with five wavelength dispersive spectrometers (WDS) and a W filament electron source. Measurements were done with 15 kV high voltage, 200 nA probe current and 25 μm beam size. The following reference materials were used for the quantification: Al – H103 Anorthite, Si – H118 Albite Piz Beverin, K – H120 Sanidine Madagascar, Ti – D015 Rutile (synthetic), Ca – H103 Anorthite, Cr – D028 Chromite, Fe – SM Fayalite, Mn – D023 Pyrolusite (synthetic), Na – H118 Albite Piz Beverin, Mg – H083 Forsterite (synthetic). Background was corrected using the mean atomic number (MAN) correction on the reference materials. Mapping was done at 40 ms dwell time and an image resolution of 729×1157 pixels, and the resulting mineral distribution maps were processed using the CalcImage software.

Bulk-rock geochemistry

Bulk-rock compositions of fresh and representative samples were measured by X-ray fluorescence (XRF) at ETH Zurich. The samples were trimmed with a diamond rock saw and homogeneous portions (150–300 g) were crushed in a hydraulic press to <4 mm grain size. The material was milled to powder in a tungsten carbide vibratory disc mill and dried at 110 $^{\circ}\text{C}$ for 12 h. Around 0.8 to 1.6 g of dry sample powder was oven-fired in an alumina crucible for 2 h at 950 $^{\circ}\text{C}$ to remove bonded volatile components, and the loss on ignition (LOI) was determined by weighing the powder before and after firing. The fired powder was then mixed with lithium tetraborate–lithium metaborate (66%/34%) fluxing agent in a 1:6 to 1:10 ratio depending on the fusibility of the sample, homogenized in an agate mortar, and cast into glass disks using a PANalytical Claisse Eagon 2 automated fusion instrument. The glass disks were analyzed with a PANalytical Axios wavelength dispersive XRF spectrometer (WD-XRF, 2.4 kV) at ETH Zurich for major and minor element compositions, as well as selected trace elements ([Supplementary Table S2](#)).

Zircon separation, visualization, U–Pb dating and trace element analysis by LA-ICP-MS

Zircon crystals were separated from the rock samples at ETH Zurich using standard techniques including Selfrag fragmentation, methylene iodide heavy liquid separation and hand-picking under a binocular microscope. All zircons were thermally annealed for 48 h at 900 $^{\circ}\text{C}$ and mounted in Struers EpoFix resin. The epoxy mounts were ground with silicon carbide polishing

paper to expose the zircon longitudinal section, then polished with diamond suspension on polishing pads to a 0.25 μm finish and coated with 20 nm carbon. Cathodoluminescence (CL) images of the zircon grains were obtained at ETH Zurich to reveal the internal zonation, inclusions, or inherited cores, using a JEOL JSM-6390 laser ablation (LA) scanning electron microscope equipped with a Deben Centaurus panchromatic CL detector.

Uranium-lead isotope and trace element compositions of zircon were measured at ETH Zurich with an ASI (Resonetics) RESolution S-155193 nm ArF excimer LA system equipped with a Laurin Technic dual-volume ablation cell at a laser repetition rate of 4 Hz. On-sample fluence was $\sim 2 \text{ J cm}^{-2}$, sample spot size was $\sim 19 \mu\text{m}$, and each point was ablated for 30 s. High-purity helium was used as carrier gas to transport the sample aerosol, and Ar was added as a make-up gas with a flow rate of 0.90–0.99 L min^{-1} . Between 4 and 129 spots were measured for each sample. More information on the analytical conditions and standardization are provided in [Supplementary Table S3](#). The data were reduced with the Iolite 4 software (Paton *et al.*, 2011) and the VizualAge add-on (Petrus & Kamber, 2012), using the model for down-hole fractionation correction of Paton *et al.* (2010). Uncertainties are stated and plotted as 2 standard errors (2 s).

Calculation and visualization of concordia ages and weighted mean ages were done with the IsoplotR tool (Vermeesch, 2018). Data points showing more than 20% of discordance (calculated from $^{207}\text{Pb}/^{235}\text{U}$ vs $^{206}\text{Pb}/^{238}\text{U}$ ratios) were rejected from the weighted mean calculations. Based on average trace element compositions of magmatic zircon crystals (Trail *et al.*, 2017), data points showing $>35 \mu\text{g/g}$ Al and/or $>2000 \mu\text{g/g}$ P were interpreted as being affected by silicate/melt or apatite inclusions, respectively, and discarded from the U–Pb and trace element databases. Grains with unusually high LREE contents indicative of a hydrothermal origin and/or apatite contamination (Hoskin & Schaltegger, 2003; Hoskin, 2005; Zhong *et al.*, 2018), and grains with $>30 \mu\text{g/g}$ Ti, indicative of rutile or Fe–Ti oxide inclusions (e.g. ilmenite), were also excluded from the age calculation and trace element database (data marked in red; [Supplementary Table S3](#)).

Zircon Hf isotope compositions by LA-MC-ICP-MS

Measurements of hafnium isotope compositions were done on the same spots as for LA-ICP-MS U–Pb dating and performed on a Nu Plasma 2 multi-collector inductively coupled plasma mass spectrometer (MC-ICP-MS) at ETH Zurich coupled to an ASI (Resonetics) RESolution S-155193 nm ArF excimer LA system. The measurements were done with a spot size of $\sim 50 \mu\text{m}$, a repetition rate of 5 Hz and an on-sample fluence of $\sim 4 \text{ J cm}^{-2}$, with high-purity He and Ar as a carrier gas to transport the sample aerosol to the spectrometer. Raw data were processed using the Iolite software following an in-house data reduction scheme. The following zircon reference materials were measured for testing the accuracy and external reproducibility: GHR 1 (Eddy *et al.*, 2019), RAK-17 (Webb *et al.*, 2020), 91500 (Wiedenbeck *et al.*, 1995, 2004), GJ-1 (Morel *et al.*, 2008), Mud Tank (Gain *et al.*, 2019) and Plesovice (Sláma *et al.*, 2008). Isobaric interferences of ^{176}Yb and ^{176}Lu on ^{176}Hf were corrected based on measured ^{175}Lu and ^{173}Yb and for $^{176}\text{Yb}/^{177}\text{Hf}$ ratios of up to 0.1. Analyses with higher $^{176}\text{Yb}/^{177}\text{Hf}$ were discarded. Further information about the Hf isotopic ratio correction is provided in [Supplementary Table S4](#). The initial Hf isotopic compositions were calculated from $^{206}\text{Pb}/^{238}\text{U}$ ages acquired by LA-ICP-MS for the same grains, the Chondritic Uniform Reservoir (CHUR) parameters of Bouvier *et al.* (2008) and the ^{176}Lu decay constant of Söderlund *et al.* (2004).

High-precision zircon geochronology by CA-ID-TIMS

High-precision zircon dating was carried out in the laboratories of ETH Zurich on concordant and inclusion-free zircon grains (3–8 grains per sample; see [Supplementary Table S5](#)), which were previously dated by LA-ICP-MS spot analyses and then plucked out from the epoxy mounts. The Northern Etendeka and quartz monzonite zircon grains used for CA-ID-TIMS dating were not mounted and analyzed by LA-ICP-MS, because their high aspect ratio would have prohibited removal from the epoxy mounts. In total, 35 single-crystal U–Pb dates were obtained.

The thermally annealed zircon grains were transferred with a micropipette into 200- μl Savillex PFA microcapsules where they underwent chemical abrasion with concentrated HF at 190 °C for 12 h inside a steel pressure vessel (Mattinson, 2005). The abraded grains were then subjected to two cleaning steps in 3-ml Savillex PFA hex beakers with 6 N HCl and 7.5 N HNO_3 in an ultrasonic bath. After rinsing, the clean grains were loaded into 200- μl PFA microcapsules with a micropipette and spiked with 6–15 mg of the EARTHTIME ^{202}Pb – ^{205}Pb – ^{233}U – ^{235}U tracer (Condon *et al.*, 2015; McLean *et al.*, 2015). Dissolution of the material was achieved in a steel pressure vessel with 40 μl of concentrated HF at 210 °C for at least 60 h. The sample solution was evaporated on a hot plate (140 °C), and conversion of the fluoride species into chloride was done by redissolving using 40 μl 6 N HCl at 190 °C (12 h) in the steel pressure vessel. The solutions were evaporated, redissolved in 3 N HCl and loaded into 50- μl ion chromatography columns filled with Bio-Rad™ AG1-X8 resin. Matrix elements were washed out with 3 N HCl, and Pb and U were eluted into the sample beakers with 6 N HCl and water, respectively. A micro-drop of H_3PO_4 was added to the purified Pb and U solutions, which were then dried on the hot plate (140 °C).

The dried samples were loaded onto outgassed, zone-refined 99.999% Re filaments with a silica gel emitter (Gerstenberger & Haase, 1997). Uranium and Pb isotope compositions were measured at ETH Zurich with a Thermo Scientific Triton Plus thermal ionization mass spectrometer (TIMS). Lead isotopes were analyzed in static mode using Faraday cups equipped with $10^{13} \Omega$ resistors (von Quadt *et al.*, 2016), with the following species detected: ^{202}Pb (spike), ^{205}Pb (spike), ^{206}Pb , ^{207}Pb , ^{208}Pb . ^{204}Pb was collected with the axial secondary electron multiplier. Mass fractionation was corrected online using the $^{202}\text{Pb}/^{205}\text{Pb}$ ratio of the tracer. Uranium was detected in static mode as UO_2 with Faraday cups equipped with $10^{13} \Omega$ resistors. The following U species were detected: $^{233}\text{U}^{16}\text{O}_2$ (spike; mass 265), $^{235}\text{U}^{16}\text{O}_2$ (mass 267), $^{238}\text{U}^{16}\text{O}_2$ (mass 270), $^{238}\text{U}^{16}\text{O}^{18}\text{O} + ^{238}\text{U}^{18}\text{O}^{16}\text{O}$ (mass 272). Correction of oxide interferences was done online using the $^{18}\text{O}/^{16}\text{O}$ ratios obtained by the analysis of mass 272 (Wotzlav *et al.*, 2017), or, in the case of low signal intensities, a fixed ratio $^{18}\text{O}/^{16}\text{O} = 0.002064 \pm 20$ was used. Instrumental mass fractionation correction was carried out offline using the isotopic composition of the double spike and assumed sample $^{238}\text{U}/^{235}\text{U}$ of Hiess *et al.* (2012). Data reduction and age calculations were done with TRIPOLI and ET_Redux (Bowring *et al.*, 2011; McLean *et al.*, 2011) using decay constants of Jaffey *et al.* (1971), and a fixed Th/U_{magma} ratio of 3.5 was used to correct for initial Th/U disequilibrium. The composition of the Pb blank was $^{206}\text{Pb}/^{204}\text{Pb} = 18.40 \pm 0.58$, $^{207}\text{Pb}/^{204}\text{Pb} = 15.41 \pm 0.55$, and $^{208}\text{Pb}/^{204}\text{Pb} = 37.0 \pm 1.3$. Uranium blank concentrations were estimated as $0.07 \pm 0.01 \text{ pg}$ for samples processed prior to February 2023 (RRBR1904, YSBR2202, YSBR2204), and as $0.27 \pm 0.06 \text{ pg}$ for all remaining samples based on the monitoring

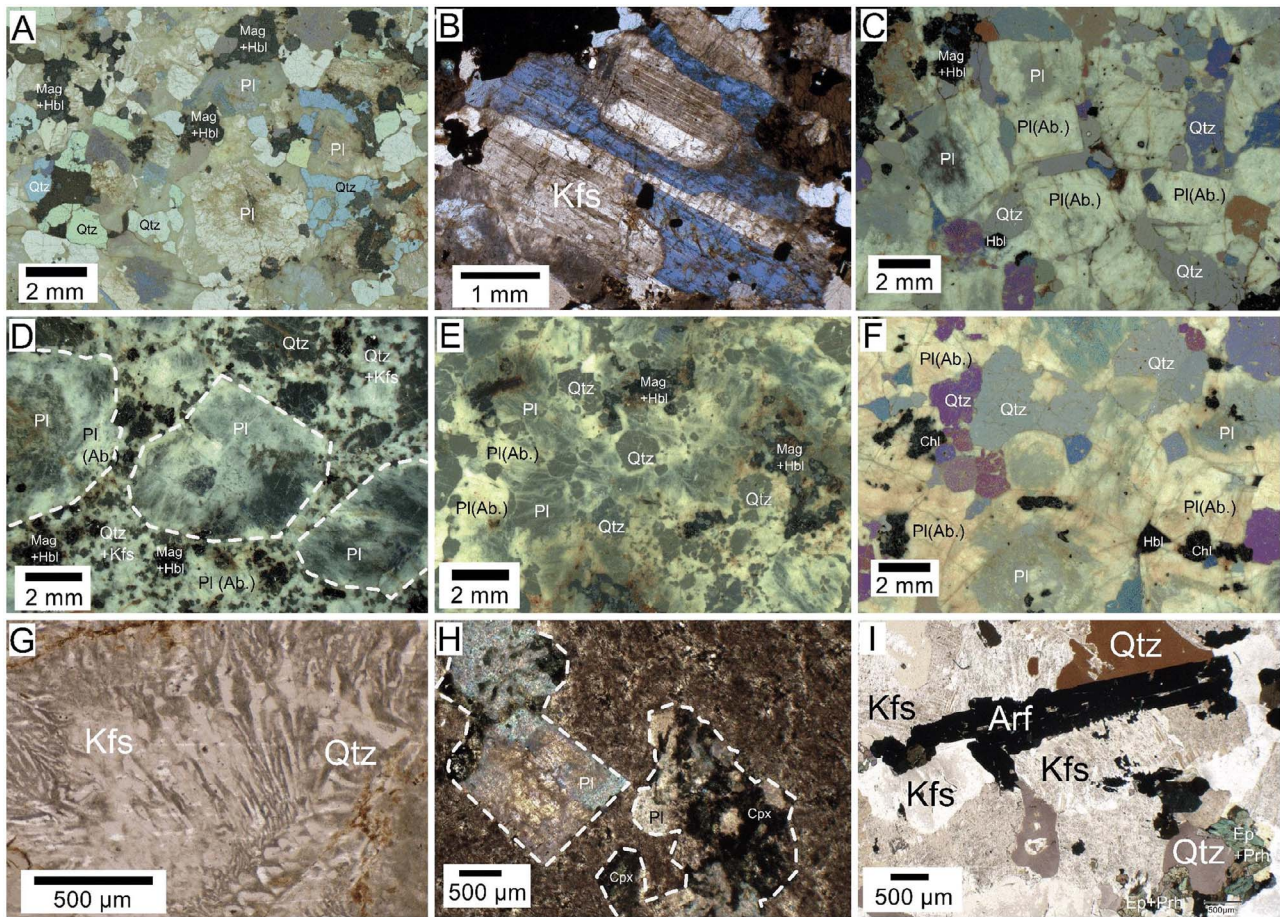


Fig. 4. Thin section photographs showing the mineralogy and textures of Brandberg igneous rocks. (A) Melanocratic quartz monzonite (YSBR2213) from the rim breccia zone; (B) perthitic feldspar crystal from the rim breccia quartz monzonite (YSBR2213); (C–D) equigranular to porphyritic edenite-augite granite (YSBR2202–YSBR2203); (E) yellowish ring dyke granite with seriate texture; (F) edenite-biotite granite (YSBR2204); (G) granophyric K-feldspar/quartz intergrowths in the Naib quartz monzonite (B102); (H) porphyritic trachydacite with glomerocrysts of plagioclase and augite; (I) peralkaline granite (RRBR1904) from the Amis complex. Mag—magnetite, Hbl—hornblende, Pl—plagioclase, Ab—albite-rich plagioclase, Qtz—quartz, Kfs—K-feldspar, Arf—arfvedsonite, Ep—epidote, Prh—prehnite.

of total procedural blanks. Bayesian eruption ages for the volcanic rocks were calculated using the BayeZirChron.c tool by Keller *et al.* (2018). All age uncertainties are reported and visualized as 95% confidence intervals.

RESULTS

Petrography

Brandberg

The Brandberg pluton consists of at least four distinct magmatic phases, which can be recognized based on field relationships and petrography (Fig. 1B and C). No volcanic rocks unequivocally attributed to the early caldera collapse (phase 1) that formed the peripheral ring fault were found during our sampling. The main granite units (phase 2a–d; Fig. 4A–F) are stratigraphically the oldest Brandberg lithologies preserved today and display a concentric zonation from the rim towards the centre. At the entrance to the Naib gorge at the NW rim of Brandberg, a quartz monzonite is observed within the outermost breccia zone (phase 2a) and intruding into the surrounding metapelitic and volcanic country rocks, with signs of magmatic stoping (Fig. 2C). This rim breccia quartz monzonite (Fig. 4A and B) is melanocratic, medium-to coarse-grained, and shows a smoky brown colouration. The rock is slightly porphyritic in texture and contains subhedral, dark

brown plagioclase grains (up to 5 mm) with thin albitic margins (<1 mm) (Fig. 4B), and smaller, subhedral quartz grains (up to 2 mm). Mafic minerals (hornblende, subordinate biotite, Fe–Ti oxides) and apatite form abundant crystal clumps, which can reach up to 4 mm in size.

Following the outer rim breccia, an edenite-augite granite (phase 2b) was emplaced and crops out in a ring extending 1–5 km into the interior of the pluton. This grey to pink, medium- to coarse-grained granite ranges from equigranular to porphyritic in texture and locally contains mafic enclaves in the cm to dm size range surrounded by reaction rims of microgranite (Fig. 2D). The edenite-augite granite contains plagioclase, K-feldspar, quartz, amphibole (commonly with relict cores of clinopyroxene), biotite, chlorite and Fe–Ti oxides (Fig. 4C and D). Accessory phases include apatite, zircon, titanite, chevkinite and locally fluorite. Blocky, euhedral to subhedral plagioclase grains (ca. 0.5–1 cm) are rimmed by white to pinkish perthite or pure albite. K-feldspar and quartz are interstitial phases and show micrographic intergrowths (Fig. 4D). Subhedral, locally poikilitic hornblende (edenite) forms brownish grains up to 3 mm in size, typically in close association with the other mafic minerals (interstitial biotite and Fe–Ti oxides) and apatite.

Between the edenite-augite granite ring and the central intrusions of the complex, a yellow-coloured granite (phase 2c; Fig. 4E)

is locally present as a discontinuous, narrow (<1 km) ring dyke. This ring dyke granite is characterized by a seriate fabric of quartz crystals, and by large (5–10 mm) crystal clumps of mafic minerals (Fig. 4E). Plagioclase is abundant as anhedral grains (3–5 mm) with more sodic/albitized margins. Quartz forms rounded individual grains over a variable size range of <1 to ~3 mm, or occurs as a part of micrographic intergrowths together with alkali feldspar. The clumps include brown- to dark green, poikilitic hornblende, abundant Fe–Ti oxides, biotite and apatite. At the outcrop in the Umoab gorge, the ring dyke granite is closely associated with grey, fine-grained aplite dykes. The aplite is a leucocratic, equigranular rock consisting of white to grey plagioclase, abundant granophyric aggregates of quartz and grey alkali feldspar, and subhedral grains of biotite.

The central part of the Brandberg complex is an amalgamation of several biotite-rich granite units (phase 2d; Fig. 4F), of which only the largest and outermost edenite–biotite granite unit was sampled for this study. The edenite–biotite granite is coarse-grained, pink in colour and contains abundant miarolitic cavities from mm to cm size. The equigranular rock (Fig. 4F) consists of subhedral, light pinkish or white plagioclase grains with transparent cores (up to 5 mm) and thick (1–2 mm) margins of pink perthitic feldspar, as well as anhedral quartz with similar grain size (up to ~5 mm). Mafic minerals (hornblende, chloritized biotite) occur sparsely as interstitial aggregates (1–3 mm) or as smaller, isolated inclusions in the plagioclase.

A quartz monzonite intrusion (phase 3a) is exposed at the highest elevations in the Naib gorge as an isolated, circular plug. This quartz monzonite is medium- to coarse-grained, porphyritic and melanocratic. Plagioclase phenocrysts up to 1 cm in size are rimmed by fine-grained, micrographic aggregates of alkali feldspar and quartz (Fig. 4G). Aggregates of anhedral, dark green clinopyroxene, commonly altered to chlorite, are intergrown with Fe–Ti oxides, apatite, and biotite. Accessory zircon typically occurs as long-prismatic crystals included in biotite or other minerals. A chemically similar rock showing clearly volcanic textures (Fig. 4H) occurs in the form of partially developed ring dykes (phase 3b) outside the NE periphery of the Brandberg pluton. These dykes consist of a porphyritic trachydacite that resemble the porphyritic varieties of the Etendeka quartz latites in appearance. The dark-grey, fine-grained groundmass contains ~10% phenocrysts (plagioclase, augite) up to ~5 mm in size. Plagioclase forms blocky, euhedral crystals showing blueish iridescence due to the presence of microscopic exsolution lamellae (Fig. 4H). Augite is subhedral and is mostly intergrown with plagioclase to form crystal clumps. Pyrite is a common accessory mineral and forms euhedral crystals up to 0.5 mm in size, which are scattered throughout the matrix.

Peralkaline granites and pegmatites (phase 4) occur only in the Amis layered complex at the SW periphery of Brandberg. The granite is coarse-grained, equigranular and contains subhedral K-feldspar, quartz, arfvedsonite, prehnite and epidote (Fig. 4I). Dark blue arfvedsonite occurs throughout the rock as randomly orientated, long-prismatic crystals. Prehnite and epidote form symplectic aggregates of up to 5 mm in size. Locally, the peralkaline granite grades into pegmatite. This pegmatite contains graphic intergrowths of pink K-feldspar with quartz, and large euhedral prisms of dark green aegirine (up to 1 cm), which are commonly rimmed by golden platelets of astrophyllite. Euhedral, bipyramidal crystals of zircon (brown to green, up to 2 mm) and pyrochlore (red brown, up to 0.5 mm) appear as accessory minerals.

Northern Etendeka quartz latites

The Northern Etendeka quartz latites are in general massive, highly welded, and mostly featureless rheomorphic ignimbrites. The groundmass of these volcanic rocks was originally glassy and is typically devitrified. Locally, the units preserve signs of flow banding, strongly flattened fiamme (Fig. 5A), or rounded gas vesicles filled with late-hydrothermal minerals (calcite, celadonite, zeolites). Phenocryst contents are low (0–20%) and mineral assemblages include only 5–6 phases: plagioclase, augite, pigeonite, magnetite, apatite, and minor zircon. The Sarusas quartz latite is crystal-poor, with a phenocryst content of 5–10%. Elongated to blocky, euhedral plagioclases commonly show albite twinning and abundant melt inclusions. They occur as phenocrysts (2–3 mm) or micro-phenocrysts (0.2–1 mm). In some samples, plagioclase phenocrysts show signs of pervasive, post-magmatic alteration with near-total replacement by phyllosilicates. Titanium-rich magnetite forms micro-phenocrysts (up to ~350 μm) with exsolution lamellae of ilmenite, rounded edges and commonly occur as aggregates (glomerocrysts) with rounded apatite grains of up to ~100 μm in size (Fig. 5B and C). Pigeonite occurs sparsely as resorbed, un-zoned micro-phenocrysts up to ~350 μm in size (Fig. 5B and C). The fine-grained (<100 μm) matrix consists of subhedral grains of plagioclase, K-feldspar, clinopyroxene, Fe–Ti oxides and minor interstitial quartz.

The Khoraseb quartz latite is mineralogically similar to the Sarusas quartz latite. The rock is moderately crystal-rich (~20%) and contains large, euhedral plagioclase phenocrysts (up to 1 cm) twinned after the albite or Carlsbad law (Fig. 5D). Other phenocryst phases include unzoned augite (up to 500 μm), pigeonite (up to ~350 μm), apatite (500 μm) and titanium-rich magnetite (up to ~250 μm) with exsolution lamellae of ilmenite. The phenocryst phases typically form glomerocrysts, and, except for plagioclase and apatite, they are characterized by the presence of resorbed rims. The fine-grained, devitrified matrix consists of lath-shaped K-feldspar crystals (up to 100 μm), interstitial quartz, and finely disseminated, subhedral grains of Fe–Ti oxides (up to ~30 μm). Both quartz latites contain zircon as an accessory mineral.

Bulk-rock and plagioclase geochemistry

The Brandberg main granites have SiO_2 contents ranging from 69–73 wt % (Fig. 3A; Supplementary Table S3). The more primitive quartz monzonites and trachydacites have SiO_2 contents ranging from 61–68 wt %. The peralkaline granites of the Amis complex and minor leucogranite/aplite dykes associated with the ring dyke intrusion (phase 2c) are more silicic than the main granites with SiO_2 contents from 72 to 77 wt %. Chondrite-normalized REE patterns for Brandberg main granites show negative slopes, strongly negative Eu anomalies, and LREE concentrations enriched to up to 200–300 times the chondritic value (Fig. 7A). The Amis peralkaline granite shows an extreme range of REE with enrichments of up to 1000 times the chondritic value, and flat REE patterns with the chondrite normalized ratio La_N/Yb_N close to 1. Brandberg granite plagioclase crystals contain patchy albitic domains (Ab_{90-100}), commonly forming thick margins surrounding the more transparent core domains, and surrounded by exsolved microperthitic feldspars (Fig. 8B). The core zones show average anorthite contents of 20–30% (Fig. 8C). The highest anorthite content was found in plagioclase from the quartz monzonite, where core compositions reach An_{45} (Fig. 8C).

The Northern Etendeka quartz latites have SiO_2 contents ranging from 63 to 69 wt %. Their chondrite-normalized REE patterns show negative slopes, only minor negative Eu anomalies, and LREE

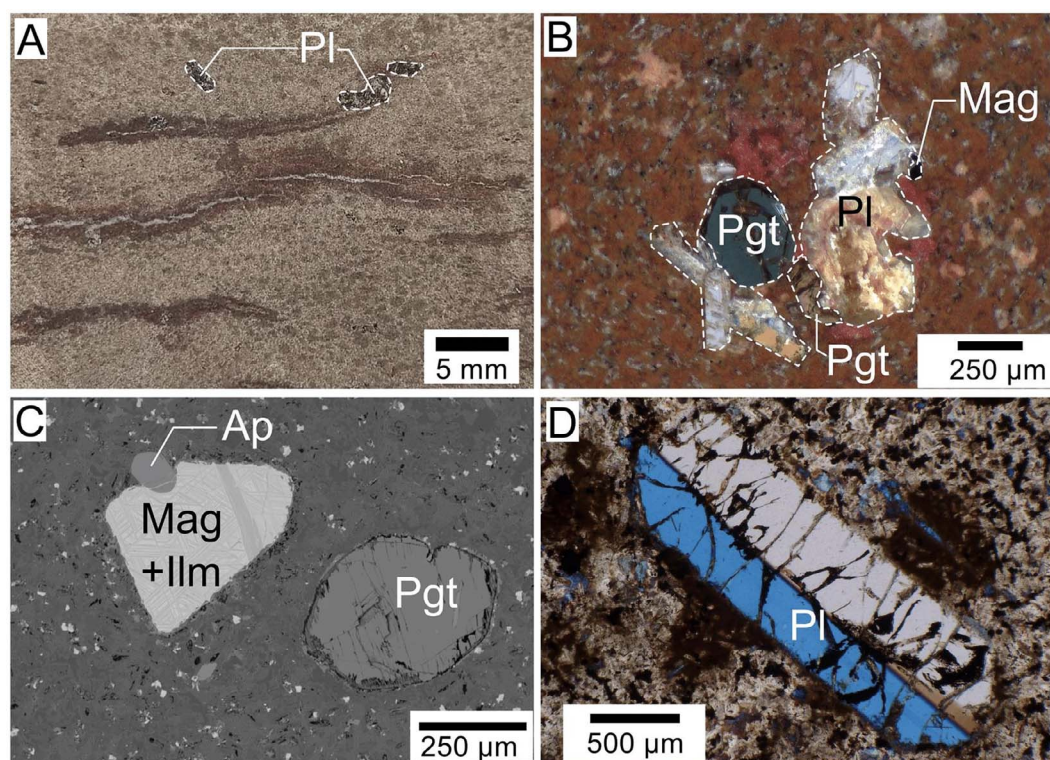


Fig. 5. Thin section photographs showing the mineralogy and textures of the Northern Etendeka volcanic rocks. (A) Crystal-poor, highly welded Sarusas quartz latite (YSSC2208) with a few plagioclase (Pl) phenocrysts and fiamme; (B) glomerocryst in the Sarusas quartz latite (YSSC2207) with plagioclase (Pl), magnetite (Mag) and pigeonite (Pgt); (C) backscattered electron image of rounded phenocrysts of apatite (Ap), magnetite with exsolutions of lamellar ilmenite (Mag + Ilm), and pigeonite (Pgt) in the Sarusas quartz latite; (D) twinned plagioclase phenocryst in Khoraseb quartz latite (YSSC2211).

concentrations are enriched to up to 200–400 times the chondritic value (Fig. 7B). Plagioclase phenocrysts in the Sarusas and Khoraseb quartz latites are unzoned and show average anorthite contents of 40–46% (Fig. S1).

The Northern Etendeka quartz latites show slightly lower SiO_2 contents compared to the Brandberg main granites, but show similar values to the more primitive Brandberg units such as the trachydacites. The chondrite-normalized REE patterns are similar for both the Northern Etendeka quartz latite and the Brandberg plutonic rocks (excluding the peralkaline Amis granites), albeit with less pronounced Eu anomalies for the former group. Across the Etendeka province, mafic and silicic volcanic rocks display a wide range of chemical compositions typical for magmatic settings driven by AFC processes (Fig. 6). Variation diagrams of minor and trace elements show the differences between the high-Ti and low-Ti suites of Etendeka volcanic rocks, and the differentiation trends for the high-Ti suites indicates particularly strongly that fractional crystallization of Fe–Ti oxides, apatite and zircon has influenced the chemical evolution of these melts (Fig. 6). Plagioclase compositions are distinct in the Brandberg plutonic suite and all Etendeka silicic volcanic units. They are less anorthitic in the plutonic rocks (ca. An_{20-30}) compared to the quartz latites (ca. An_{40-46}), although the maximum anorthite content in the most primitive Brandberg unit (quartz monzonite; An_{40}) overlaps with the plagioclase compositions of the quartz latites. No samples from the Southern Etendeka were analyzed in this study, but literature data show that the phenocrysts from the low-Ti silicic Etendeka units (e.g. the Lower Coastal quartz latites) show even higher An contents (53–65%; Ewart *et al.*, 1998b, 2004b) compared to both the Brandberg granites and the Northern Etendeka quartz latites.

Zircon U–Pb geochronology

Zircon crystals from the Brandberg main granite body range from colourless to pale yellow in colour. They are ca. 80–160 μm in size, short-prismatic and doubly terminated (Fig. 9A and B; Supplementary Figs S2–S8). Inclusions of apatite, Fe–Ti oxides or K-feldspar are present in some grains. In the Amis peralkaline granite, zircon forms colourless to brown, euhedral crystals 80–200 μm in size with bipyramidal shape (Fig. 9C and D). In the quartz monzonite, zircon crystals are thin and long-prismatic in habit, and only a few were recovered as fragments up to 150 μm in length (Fig. 9E–F). Zircon data from the edenite-biotite granite (YSBR2204) and the Amis peralkaline granite (RRBR1904) were already published in Sun *et al.* (2024) and are only reported for completeness.

Detrital zircons collected from dry river sand at the Naib and Tsisab gorges (Fig. 1B) gave overlapping weighted mean U–Pb ages of 132 ± 2 Ma (YSBR2206; $n=94$, $\text{MSWD}=1.3$) and 133 ± 2 Ma, respectively (YSBR2211; $n=70$, $\text{MSWD}=0.97$; Supplementary S12A and B). Both detritus samples contain ~6% older grains showing Late Palaeozoic to Neoproterozoic ages (479 to 647 Ma), which represent either inherited crystals or, less likely due to the sampling location inside the Brandberg complex, detrital grains derived from granitic country rocks. The low MSWD values for the detrital samples confirm a short lifespan for the whole complex and suggest that the intrusion of the central parts of the pluton, which were not covered by rock samples in this study, was simultaneous with the outer parts within the uncertainty of LA-ICP-MS measurements. All ages described and discussed in the further paper refer to zircon separated from rock samples. Weighted mean zircon $^{206}\text{Pb}/^{238}\text{U}$ ages for the Brandberg main granite body obtained by LA-ICP-MS (Supplementary Figs S10–S12; Table S3)

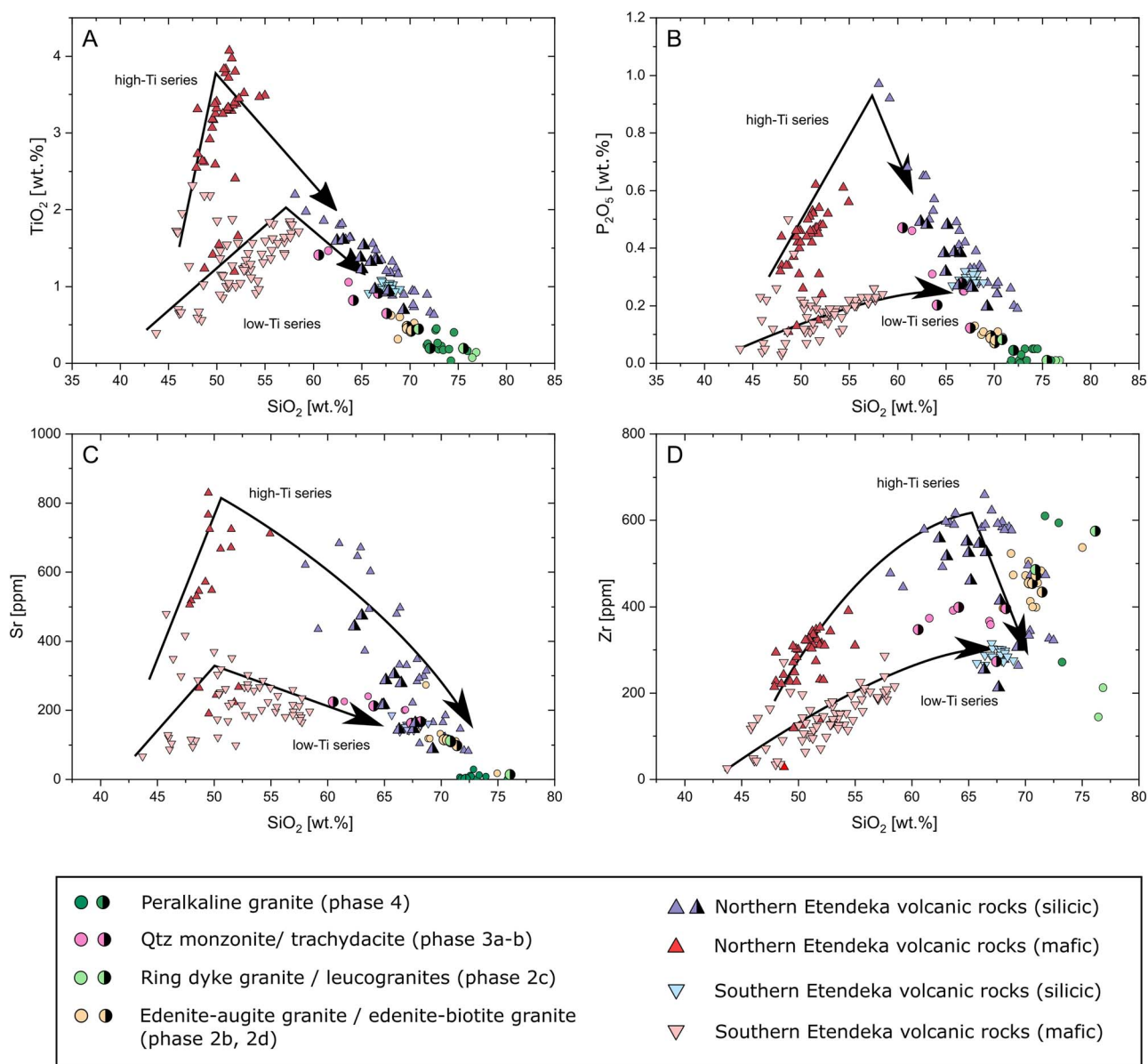


Fig. 6. Harker diagrams for Brandberg plutonic rocks and Etendeka volcanic rocks showing the magmatic evolution of the two suites for major/minor and trace elements, including (A) titanium, (B) phosphorus, (C) strontium and (D) zirconium. Data sources as for Fig. 3.

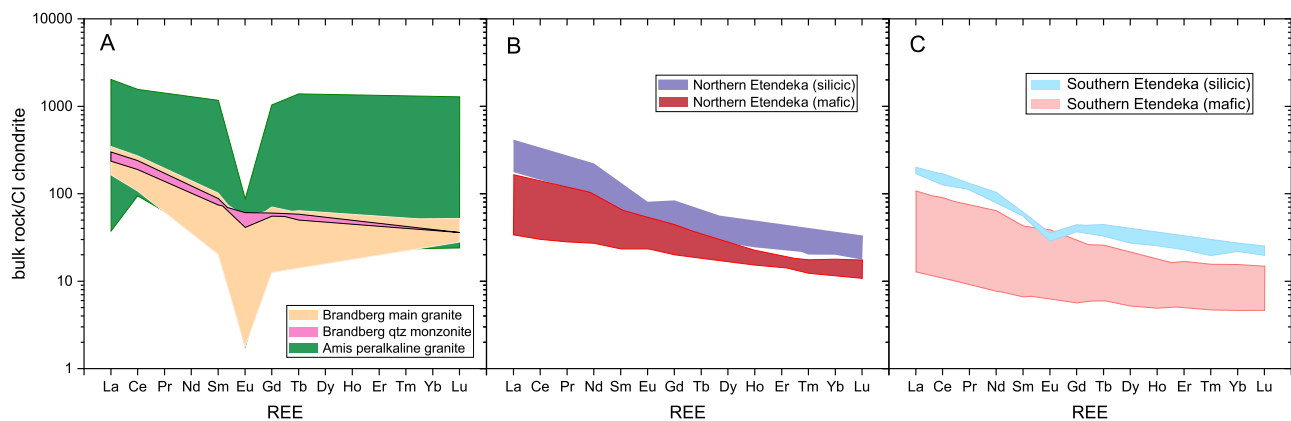


Fig. 7. Chondrite-normalized REE patterns for (A) Brandberg plutonic rocks, (B) Northern Etendeka volcanic rocks and (C) Southern Etendeka volcanic rocks. Chondrite normalizing factors from Sun & McDonough (1989).

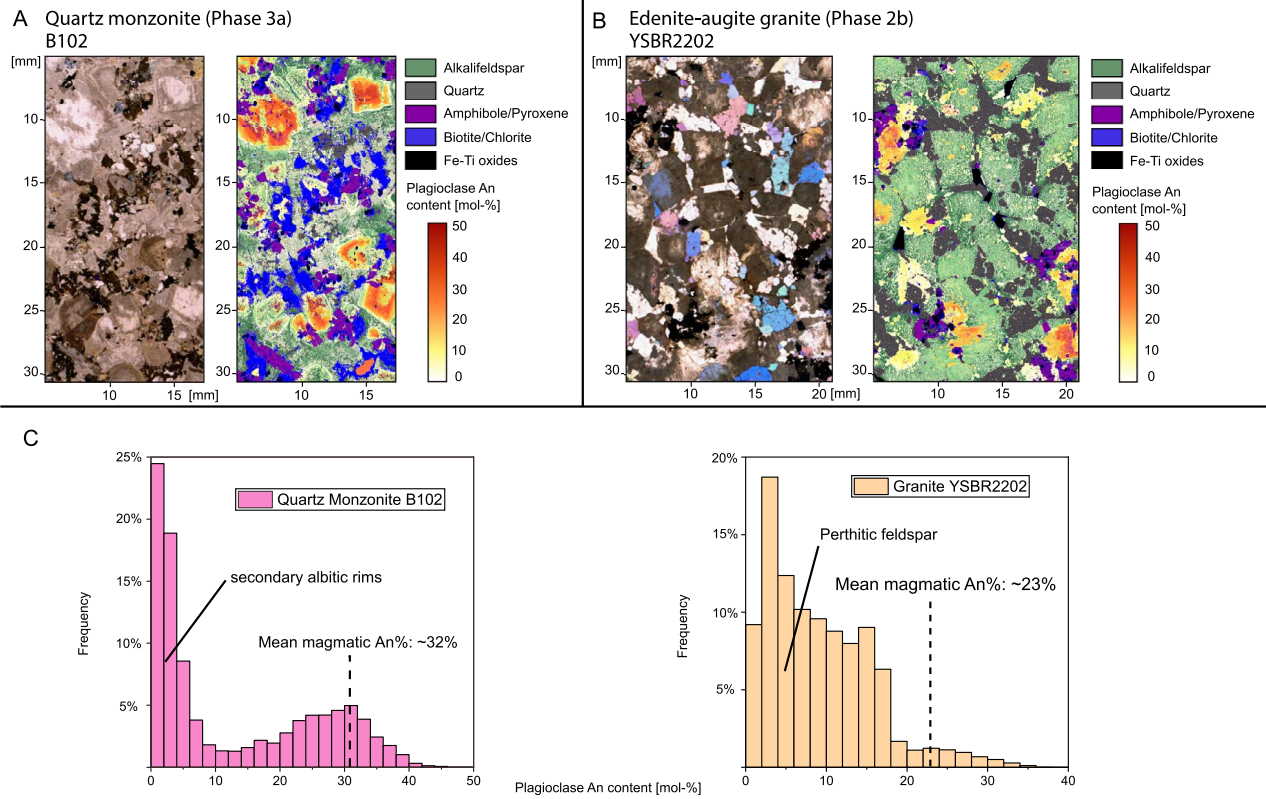


Fig. 8. Phase and plagioclase compositional maps for the (A) Naib quartz monzonite and (B) main granite derived from quantitative spectral imaging with the EPMA. (C) Histograms showing the range and abundance of anorthite contents (An) in Brandberg plagioclase.

range from 135 ± 2 Ma (BR2203; $n = 32$, MSWD = 0.51) to 131 ± 2 Ma (BR2204; $n = 34$, MSWD = 0.38). Xenocrysts and inherited cores are rare (<4%) among the analyzed zircon grains.

Single-crystal zircon dates for the Brandberg intrusive units determined by CA-ID-TIMS range from 133.20 ± 0.19 Ma (ring dyke granite, phase 2c) to 131.838 ± 0.043 Ma (peralkaline granite, phase 4; Sun et al., 2024), with low-age dispersion within each sample (Fig. 1C; Supplementary Table S5). As the effects of secondary Pb loss were minimized by the chemical abrasion procedure, most grains selected for dating (Supplementary Figs S6 and S7) are concordant and therefore these dates are interpreted as the age of zircon crystallization. The rim breccia quartz monzonite (phase 2a) yielded zircon ages from 132.856 ± 0.098 Ma to 132.81 ± 0.11 Ma. The edenite-augite granite (phase 2b), the ring dyke granite (phase 2c) and the edenite-biotite granite (phase 2d; Sun et al., 2024) are indistinguishable in age and individual zircon ages range from 133.20 ± 0.19 to 132.62 ± 0.29 Ma. An unusually old zircon crystal (133.77 ± 0.41 Ma) that shows abundant mineral inclusions was found in the ring dyke granite and it likely represents an antecryst. Zircon ages for the Naib quartz monzonite plug (phase 3a) are 132.351 ± 0.092 to 132.14 ± 0.18 Ma with a substantially older antecryst dated at 132.748 ± 0.059 Ma. These dates confirm that the monzonite intruded after the main granite body. The zircon grains from the peripheral trachydacite dykes (phase 3b) are highly metamict and two grains dissolved completely during chemical abrasion. The three surviving grains range in age from 132.733 ± 0.042 to 132.17 ± 0.51 Ma. Zircon dates from the Amis peralkaline granite (phase 4) are 131.910 ± 0.039 to 131.838 ± 0.043 Ma (Sun et al., 2024) and indistinguishable within uncertainties (Fig. 1C).

Zircon grains separated from the Northern Etendeka quartz latites are up to ~ 300 μm in length, long-prismatic, and commonly

show dark-coloured melt inclusions aligned along the central axis of the crystals (Fig. 9G). The crystals are sector or oscillatory zoned and are mostly free of inherited cores (Fig. 9H). *In situ* zircon dating by LA-ICP-MS yielded overlapping weighted mean U-Pb ages of 137 ± 2 Ma (MWSW = 0.79; $n = 10$) and 137 ± 3 (MSWD = 0.06; $n = 2$) for the Sarusas and Khoraseb quartz latites, respectively.

Zircon U-Pb dates obtained by CA-ID-TIMS (Fig. 1C; Supplementary Table S5), which were from a different set of grains than the LA-ICP-MS analyses, show that the quartz latite magmas erupted during or before the emplacement of the Brandberg pluton (Fig. 1C). Zircons from the Sarusas quartz latite yielded a Bayesian eruption age of 132.572 ± 0.034 Ma based on the six youngest zircon single-crystal dates forming a single population. Two older dates at 132.975 ± 0.061 and 133.297 ± 0.078 Ma likely record antecryst crystallization in the magma chamber and were excluded from the eruption age calculation. For the Khoraseb quartz latite, we obtained a Bayesian eruption age of 133.319 ± 0.073 based on the five youngest zircon grains. Three antecrystic zircon crystals were dated at 134.238 ± 0.075 , 133.902 ± 0.063 and 133.753 ± 0.065 . Our new geochronological data indicate that the high-Ti silicic volcanism started at least by 133.3 Ma with the eruption of the Khoraseb quartz latite and lasted at least for 0.7 Myr until the 132.6 Ma eruption of Sarusas.

Zircon geochemistry and Hf isotope compositions

Zircon grains from the Brandberg main granite body and the trachydacite dykes have similar geochemical features (Fig. 10; Supplementary Table S3). Rare earth element patterns show steep, positive slopes, positive Ce anomalies and negative Eu anomalies (Supplementary Fig. S15). Zircon Zr/Hf ratios are between 35 and 60 in all samples. The Eu anomaly $(\text{Eu}/\text{Eu}^*)_{\text{N}}$, where N signifies the

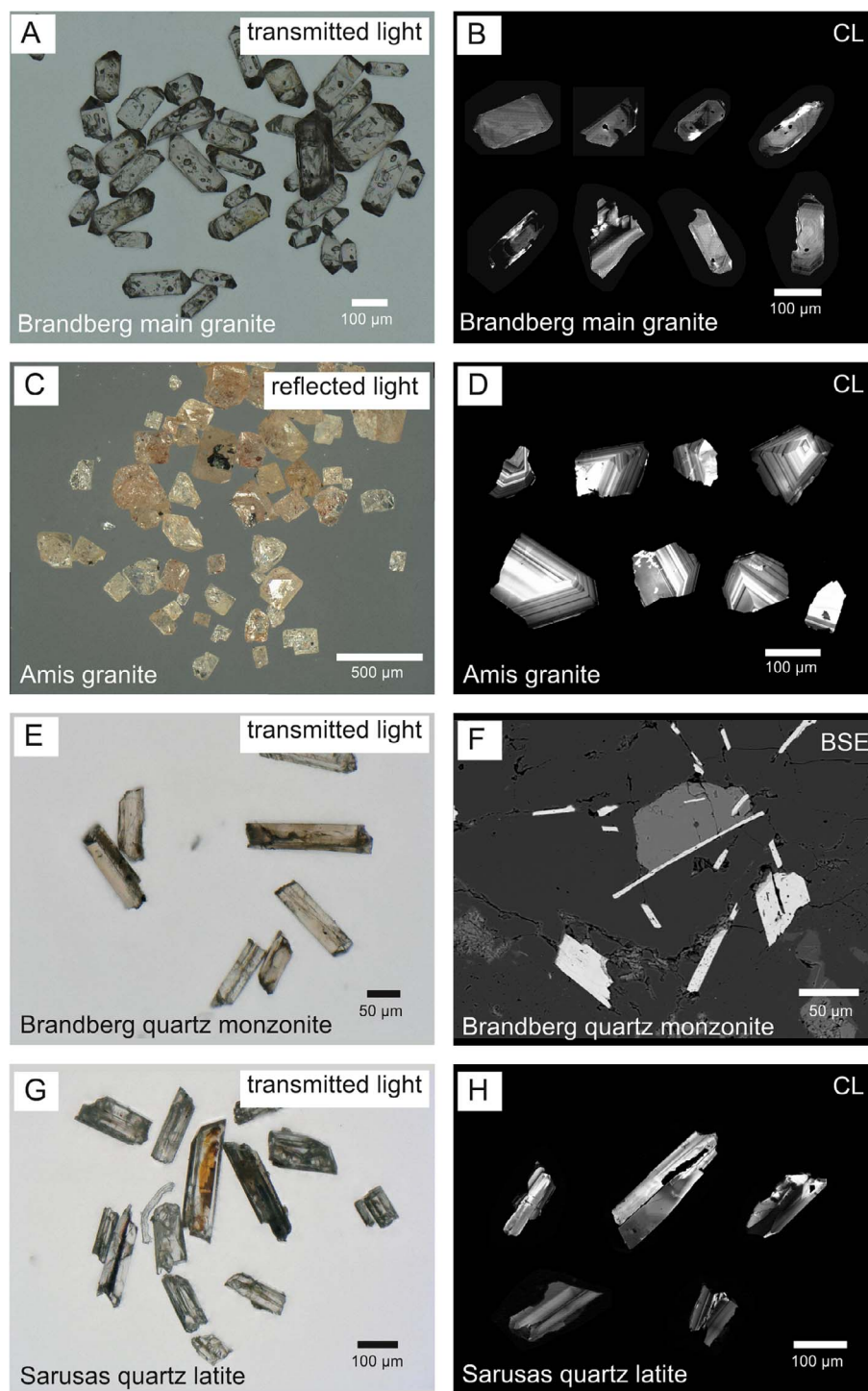


Fig. 9. Optical and cathodoluminescence (CL) images for representative zircon grains from (A–B) the Brandberg main granite, (C–D) the Amis peralkaline granite, (E–F) the Naib quartz monzonite and (G–H) the Sarusas quartz latite.

chondrite-normalized concentration, was calculated from concentrations of the two neighbouring REE using the formula:

$$Eu_N^* = \sqrt{(Sm_N \times Gd_N)}. \quad (1)$$

Values for $(Eu/Eu^*)_N$ range from ca. 0.3 to 0.02 (Tang et al., 2021). Zircons from the trachydacite dykes show highly variable REE contents with some grains being enriched in REE by a factor of ~10 relative to others, which might be due to apatite inclu-

sions. Titanium contents for zircon from the main granite and trachydacite are 4–18 $\mu\text{g/g}$ (11 on average), Th/U ratios are 0.2–1, and Yb/Gd ratios are 5–30. The zircon from the Amis peralkaline granite has high Yb/Gd ratios of 100–1000, very low Ti contents of <3 $\mu\text{g/g}$ and high Th/U of 1–4. The hafnium isotope compositions (Fig. 11) of zircon crystals from the Brandberg main granite, the ring dyke granite and a rim breccia quartz monzonite show overall variations in ϵ_{Hf} (132 Ma) from ca. –2.8 to +2.7 with average values between –0.4 and +0.5 ($n=93$).

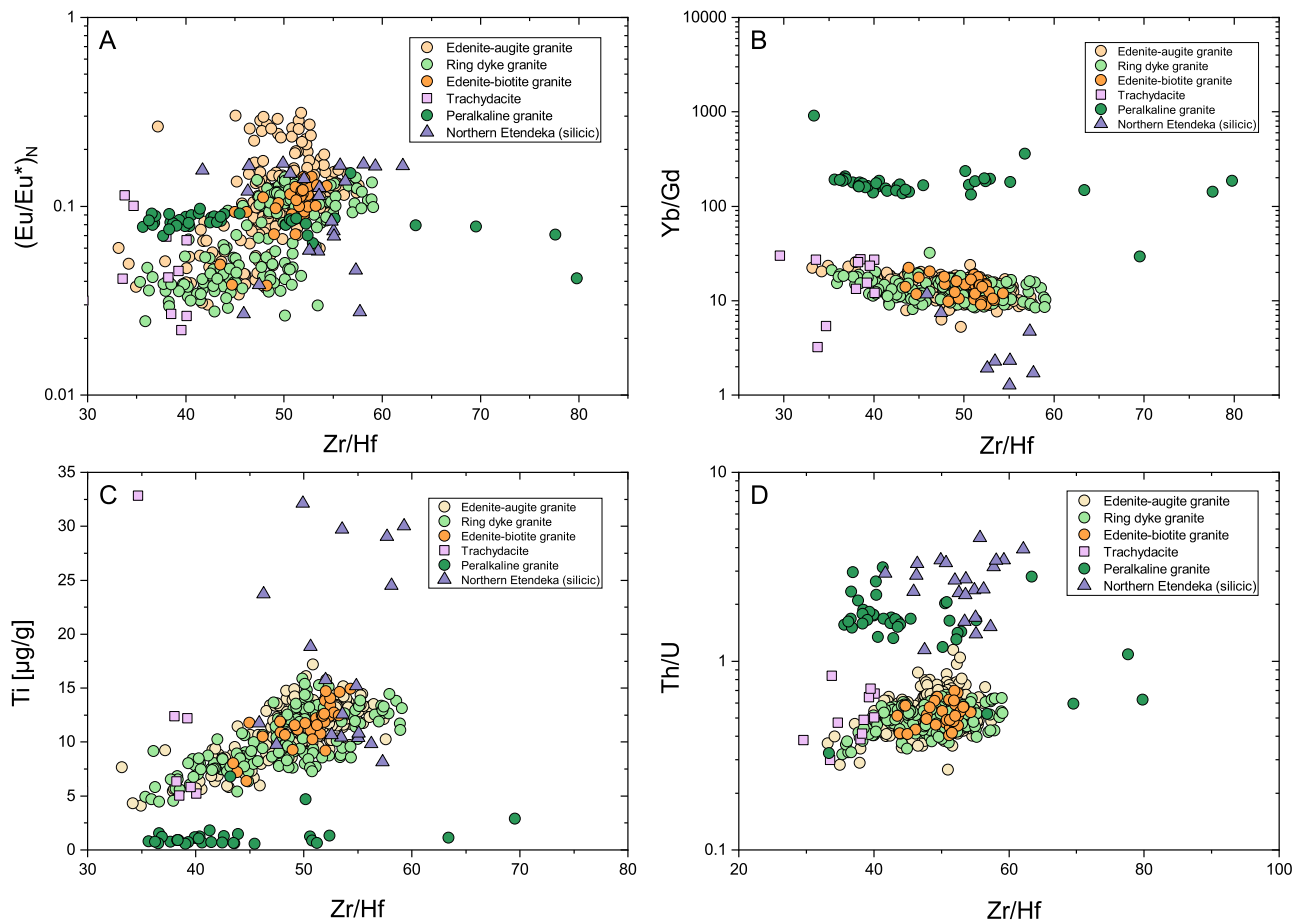


Fig. 10. Geochemical plots for Brandberg and Northern Etendeka zircon grains, including (A) Eu anomaly versus Zr/Hf, (B) Yb/Gd versus Zr/Hf, (C) titanium concentrations versus Zr/Hf and (D) Th/U versus Zr/Hf.

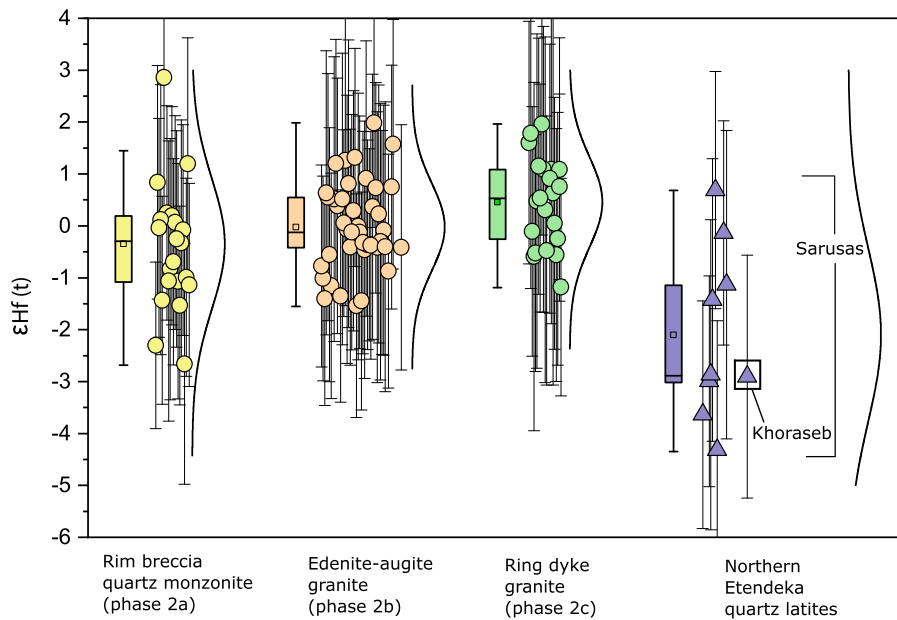


Fig. 11. Hafnium isotope compositions of Brandberg and Northern Etendeka zircon grains plotted with 2-s error bars and Kernel density plots. The line and dot in the box plot show the median and mean values for each sample, respectively.

The zircon crystals from the Northern Etendeka quartz latites show similar values of $(\text{Eu}/\text{Eu}^*)_{\text{N}}$ and Zr/Hf compared to the Brandberg main granite, but they have slightly flatter REE patterns, low Yb/Gd ratios of 0.05–12, high Ti contents of 10–51 $\mu\text{g/g}$

(average 23 $\mu\text{g/g}$), and Th/U ratios above 1 (Fig. 10). In the Sarusas quartz latite, zircon ϵHf (132 Ma) values are on average lower than for Brandberg and range between -4.3 ± 2.5 and $+0.7 \pm 2.3$, with a mean value of -2.0 ± 0.8 ($n=9$) (Fig. 11; Supplementary Table S4).

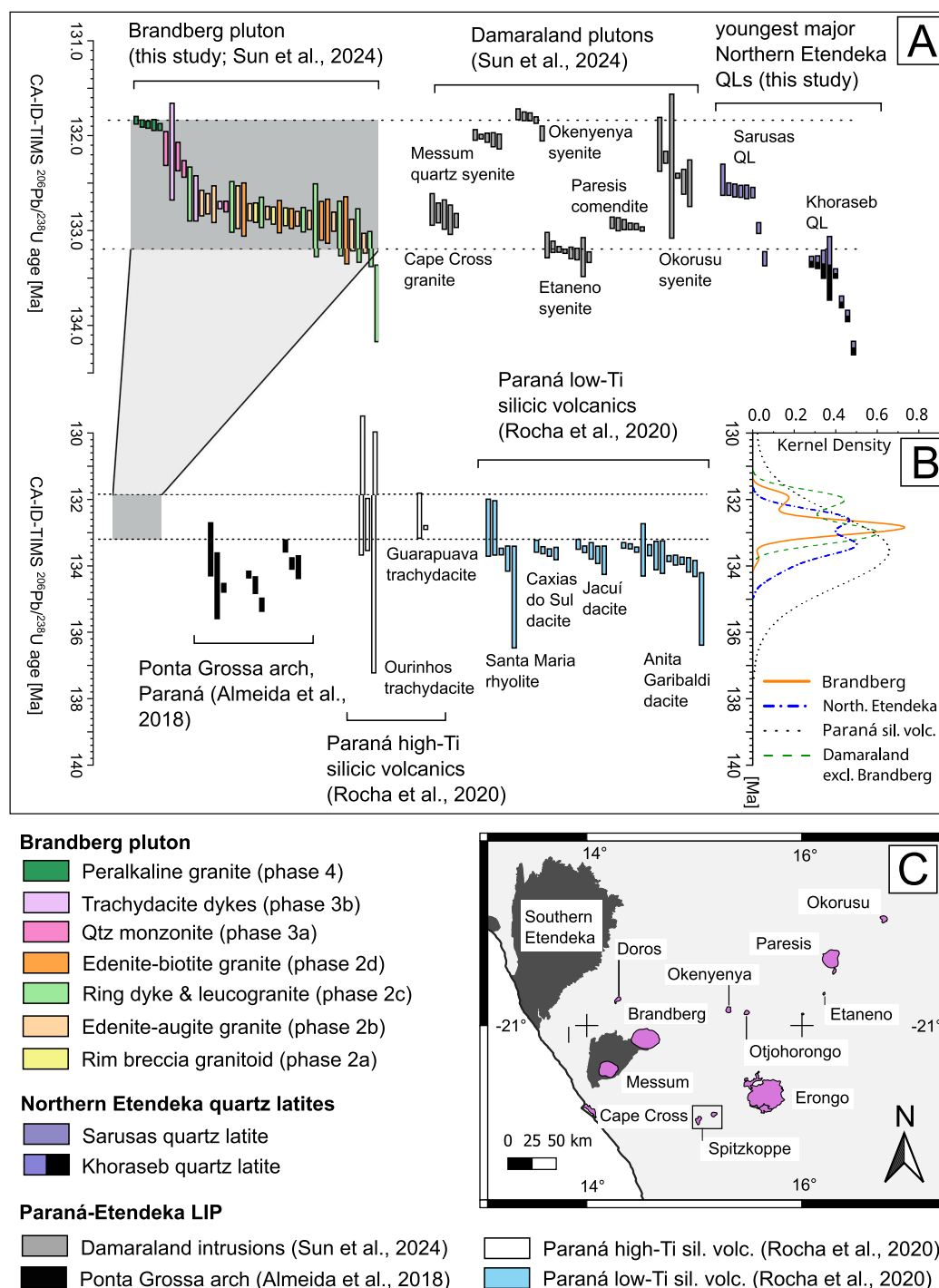


Fig. 12. (A) Rank order plot showing a compilation of available CA-ID-TIMS zircon U-Pb dates for the Brandberg pluton (this study), the Northern Etendeka quartz latites (this study), mafic to intermediate intrusive rocks of the Ponta Grossa arch (Almeida et al., 2018) and silicic volcanic rocks of the Paraná province, Brazil (Rocha et al., 2020). (B) Kernel density plot comparing the CA-ID-TIMS zircon ages of Brandberg and Northern Etendeka from this study with other CA-ID-TIMS data of plutonic and volcanic units in the Paraná-Etendeka LIP taken from the literature. Literature data are from Rocha et al. (2020) and Sun et al. (2024). (C) Map of the Damaraland province showing the locations of the intrusive complexes.

In the Khoraseb quartz latite, Hf isotope compositions could only be retrieved from a single grain due to the low abundance of zircon in the rock, and the high concentrations of Yb in the zircon grains that interferes with the Hf isotope measurements. The Khoraseb zircon grain yielded an Hf isotope composition of $\epsilon_{\text{Hf}}(132 \text{ Ma}) = -2.9 \pm 2.4$ (Fig. 11; Supplementary Table S4), which overlaps within uncertainties with the isotopic range shown by the Sarusas zircon grains.

DISCUSSION

Ages of Brandberg and the Northern Etendeka quartz latites in the framework of the Paraná-Etendeka LIP

Our new zircon U-Pb ages for the Brandberg complex show that it was emplaced within ca. 1.4 Myr during the late stages of the Paraná-Etendeka LIP, after the eruption of most quartz latites and basalts dated so far to high precision (Fig. 12A and

B). Assembly of the Brandberg complex from 133.2 to 131.8 Ma closely overlaps that of the other Damaraland complexes dated by the same method (Fig. 12A and C) whose CA-ID-TIMS weighted mean ages range from 133.219 ± 0.017 Ma for the Etaneno syenite to 131.811 ± 0.023 Ma for the Okenyenya syenite (Sun *et al.*, 2024). The Etaneno complex therefore intruded around the initial stages of the construction of Brandberg, or slightly before. Zircon dating of Paresis and Cape Cross returned weighted mean ages of 132.875 ± 0.075 Ma (Paresis comendite) and 132.858 ± 0.052 Ma (Cape Cross granite) making these two complexes coeval with the Brandberg main granite body (Sun *et al.*, 2024). The Okorusu syenite (132.223 ± 0.065 Ma) is very similar in age to the trachydacite/quartz monzonite stage of Brandberg, whereas ages for the Messum quartz syenite (132.017 ± 0.022 Ma) and the Okenyenya syenite (131.811 ± 0.023 Ma) bracket the emplacement of the Amis peralkaline granite at 131.879 ± 0.020 Ma (Sun *et al.*, 2024). Erongo is the only prominent Damaraland complex that is not yet covered by precise zircon dating. However, Wigand *et al.* (2004) reported similar in-situ SIMS zircon ages from the plutonic and volcanic silicic units of the Erongo complex (rhyodacite: 131.8 ± 1 Ma, granodiorite: 132.3 ± 1.9 Ma, granite: 130.3 ± 1.4 Ma, rhyolite: 132.6 ± 4.2 Ma).

For the Northern Etendeka, the eruption ages of the late-erupted quartz latite units Khoraseb (133.319 ± 0.073 Ma) and Sarusas (132.572 ± 0.034 Ma) clearly overlap with the emplacement ages of the Damaraland plutons. They are also comparable with available dates for stratigraphically correlated eruptive units in the Paraná province (Rocha *et al.*, 2020). In Paraná, silicic volcanism started at around 133.6 Ma with several low-Ti dacites and rhyolites (Rocha *et al.*, 2020), considered counterparts of the Southern Etendeka low-Ti latites and quartz latites. Indistinguishable eruption ages of 132.72 ± 0.76 and 132.906 ± 0.072 Ma were reported for two high-Ti trachydacites Ourinhos and Guarapuava (Rocha *et al.*, 2020). These units are interpreted as the Paraná equivalents of the Khoraseb and Sarusas quartz latites, respectively, based on stratigraphic relationships and bulk rock geochemistry (Marsh *et al.*, 2001). The comparison of zircon ages from Namibia and Brazil does not contradict this interpretation. The 132.9 ± 0.76 Ma eruption age of the high-Ti silicic volcanism in Brazil (Rocha *et al.*, 2020) overlaps with our age ranges for the Sarusas and Khoraseb quartz latites, although the time difference between the two major eruptions could not be resolved in Paraná. The Santa Maria rhyolite in Brazil yielded a weighted mean eruption age of 133.65 ± 0.10 Ma based on five zircon grains (Rocha *et al.*, 2020). This unit has previously been correlated with the Fria quartz latite in the Etendeka, which lies stratigraphically between the Khoraseb and Sarusas quartz latites (Ewart *et al.*, 2004b) and, based on our results, should have been erupted between 133.3 and 132.6 Ma. The 133.65 ± 0.10 Ma weighted mean age for Santa Maria is strongly weighted towards the most precise but older analysis; however, the ages for the two youngest grains (132.86 ± 0.83 and 132.85 ± 0.87 Ma) align with both the proposed stratigraphic correlation and our zircon ages from the Etendeka units.

Geological interpretation and intrusive history of the Brandberg pluton

Zircon geochronology and supporting evidence from petrography and field observations suggest that the Brandberg complex represents a shallow, resurgent intrusion filling space left by caldera-forming eruption(s). The complex was assembled in four phases: (1) formation of a master caldera fault and likely volcanic activity, but with no magmatic products preserved today, (2) emplacement of the main granite body, (3) emplacement of

a quartz monzonite plug and the peripheral, subvolcanic trachydacite dykes and (4) emplacement of peralkaline granites and pegmatites (Fig. 12A). The concentrically zoned main granite body can be further divided into four units (2a–2d), corresponding to the centres 3–6 of Diehl (1990); while our phase 3 combines the two intermediate units defined as centres 1 and 7 by Diehl (1990).

The oldest structure present at the Brandberg pluton based on field observations is the circular fault system (phase 1) that surrounds most of the Brandberg except for the NE boundary, where the pluton intruded into basement granites (Fig. 1B). Along this fault zone, the surrounding Etendeka volcanic layers overlying Karoo metasedimentary country rocks show a progressive inward dip towards the granite contact instead of being lifted up by the granite body (Fig. 2B), which provides a strong argument for a subsidence predating the Brandberg intrusion, most likely caused by caldera collapse. A passive emplacement of the Brandberg following an earlier caldera collapse and cauldron subsidence was first proposed by Korn & Martin (1953) based on the same arguments. In this scenario, the silicic melts that stalled in the crust to form the pluton did not forcefully intrude the country rocks by opening up new fractures, but were mostly channelled into pre-existing magma pathways (Korn & Martin, 1953).

Assuming that all zircon grains crystallized at emplacement level, the recorded plutonic activity at Brandberg started with the intrusion of the main granites (phases 2a–d) at around 133.2 ± 0.19 Ma and lasted for ~ 0.6 Myr to 132.62 ± 0.29 Ma. Magmatic activity continued from 132.351 ± 0.092 to 132.14 ± 0.018 Ma with the emplacement of the small-volume Naib quartz monzonite (phase 3a), and the peripheral trachydacite dykes (phase 3b; 132.733 ± 0.042 to 132.17 ± 0.51 Ma), and finally the emplacement of the Amis peralkaline complex from 131.910 ± 0.039 to 131.838 ± 0.043 Ma (phase 4).

In the literature, the intrusive relationship between the Brandberg quartz monzonites and trachydacites (phase 3) is controversial. Diehl (1990) interpreted the Nuwuarib/Umoab trachydacite dykes as predating the Brandberg granite, but pointed out that no direct contacts are observed between these dykes and the main granite. For the Naib quartz monzonite plug, Diehl (1990) postulated a late-stage origin based on cross-cutting relationships with the main granite. However, Schmitt *et al.* (2000) reported a fine-grained porphyritic margin of main granite at the contact with the quartz monzonite, which they interpreted as a chilled margin and suggested that the quartz monzonite predates the main granite. Our new U–Pb zircon ages suggest that the quartz monzonite plug is younger than the main granite (Fig. 1C). High elevations and poor access prevented us from evaluating the contact relationships during the field campaign, so the field relationship between these units needs to be confirmed in light of the new zircon ages in the future.

The Amis peralkaline granite (phase 4) was the last Brandberg unit to be emplaced. The close association of the granite with aplites and pegmatites (Diehl, 1990; Schmitt *et al.*, 2000) indicates that the emplacement of the Amis peralkaline granite marks the transition from the magmatic to the hydrothermal realm shortly before the magmatic system reached its complete solidification. Late-stage hydrous activity is widespread throughout the Brandberg pluton, expressed by the abundance of recrystallized, microporous feldspars with thick albitic margins (Worden *et al.*, 1990; Walker *et al.*, 1995; Fonseca Teixeira *et al.*, 2022), the local presence of granophyric textures, and high H₂O contents of up to 10.6 wt % in quartz-hosted fluid/melt inclusions from the Amis pegmatite (Schmitt *et al.*, 2002). This late episode of hydrothermal overprinting most likely stems from fluids exsolved

by the crystallizing granitic magma itself and is, therefore, to be regarded as auto-metasomatism. In fact, no rocks with low bulk-rock $\delta^{18}\text{O}$ values ($<5.5\%$) indicating the involvement of meteoric water occur at Brandberg (Trumbull *et al.*, 2004). In addition, thermochronological models based on apatite fission-track ages show continuous exhumation and cooling in the Damaraland over the Late Cretaceous and Cenozoic with no evidence for any major magmatic or metamorphic events post-dating the emplacement of the Damaraland complexes, which could have provided hydrothermal fluids (Krob *et al.*, 2020).

The exact intrusion depth of the Brandberg pluton is not well constrained as Brandberg amphiboles have compositions that are not suitable for Al-in-hornblende geobarometry (Schmitt *et al.*, 2000). There are, however, indications for a shallow intrusion at the subvolcanic level (Schmitt *et al.*, 2000). The field observations show that the Brandberg intruded into Etendeka lavas that must have been at the surface. The geometry of granite ring dikes with the perimeter breccia zone and concentric dip of lavas inward, and the frequent occurrence of miarolitic cavities in the granite also suggest a very shallow intrusion. Comparison of observed and modelled fractional crystallization paths with MELTS are consistent with low-pressure (300 MPa) conditions, and phase relationships plotted in the quartz–albite–orthoclase ternary system indicate equilibration under low pressures (Schmitt *et al.*, 2000). Also, zircon crystals in the Naib quartz monzonite show elongated habits with high aspect ratios, resembling volcanic zircons and suggesting rapid crystallization (Hoskin & Schaltegger, 2003; Qian *et al.*, 2012). Finally, the presence of a fossil hot spring system at the Brandberg complex also indicates intrusion at shallow levels of the crust (Pirajno, 1987).

The proposed geological interpretation of the Brandberg as a shallow, resurgent reservoir that was emplaced into a down-sagging structure after a caldera collapse is similar to models proposed for other Damaraland complexes such as Messum or Erongo (Ewart *et al.*, 2002; Wigand *et al.*, 2004). In fact, the geologic history of the nearby Messum complex proposed by Ewart *et al.* (2002) is very similar to that of Brandberg: (1) eruption of predominantly mafic magmas in the region, causing steep geothermal gradients; (2) local down-sagging of the thermally weakened crust following a first quartz latite (Goboboseb) eruption; (3) caldera collapse and development of steeply dipping ring faults following another quartz latite (Springbok) eruption; (4) cauldron subsidence and emplacement of granitoids accompanied by resurgence and (5) injection of late-stage peralkaline intrusions into the granitoid body.

Genetic relationship between the Brandberg pluton and the Etendeka quartz latites?

The Brandberg granite intrusion and the Sarusas/Khoraseb quartz latite ignimbrites studied in this paper formed coevally and are some of the most representative units for the two different crustal levels (upper crust and surface) exposed within the Etendeka province today. We emphasize that Brandberg cannot be the eruptive vent for any of the quartz latites preserved in the Northern Etendeka province today due to the large distance (>200 km) between the outcrop areas and the distinct bulk-rock Sr–Nd compositions (Fig. S16) between the plutonic and volcanic suites. However, a broad geochemical comparison between these units is warranted in order to provide useful information on the petrogenetic processes that generated the different silicic magmas in the Etendeka province.

Plutonic roots beneath coeval volcanic deposits may represent unerupted analogues of these volcanic rocks ('failed eruptions'),

or cumulates remaining in the magma chamber ('crystal graveyards') after extraction of eruptible melt (Bachl *et al.*, 2001; Bachmann *et al.*, 2007; Gelman *et al.*, 2014; Keller *et al.*, 2015; Lundstrom & Glazner, 2016; Florez *et al.*, 2025). In anorogenic systems like the Etendeka province where the generation of silicic magmas is mainly driven by fractional crystallization (with some crustal assimilation) of mantle-derived primary magmas (Bonin, 2007; Moya *et al.*, 2021), crystal-poor ignimbrites usually form as interstitial melts extracted from a crystal-rich ('mushy') reservoir (e.g. Bachmann & Bergantz, 2004; Huber *et al.*, 2012; Ellis *et al.*, 2014). If such melt extraction processes occur, the plutons left behind should record geochemical or textural evidence of crystal accumulation (Lee & Morton, 2015; Špillar & Dolejš, 2015; Barnes *et al.*, 2016; Fiedrich *et al.*, 2017; Cornet *et al.*, 2022). This is not observed in any of the Brandberg granite units:

- (1) The Brandberg granites, with the exception of the highly evolved, late-stage peralkaline granite and leucogranite units, show comparable degrees of magmatic differentiation compared to all Northern Etendeka quartz latites including Sarusas/Khoraseb (Fig. 6). The Brandberg granites are, if anything, slightly more evolved geochemically and tend to have lower concentrations of P and Ti, suggesting more fractionation of apatite and Fe–Ti oxides in their petrogenesis. The Brandberg granites do not show clearly elevated concentrations of compatible elements (e.g. Ca, Ba or Sr) relative to the Northern Etendeka quartz latites (Fig. 6), as would be expected if they were cumulate deposits.
- (2) While the main magmatic mineral assemblages are similar in the Brandberg granites and the Sarusas/Khoraseb quartz latites (plagioclase, K-feldspar, quartz, clinopyroxene and Fe–Ti oxides), plagioclase compositions are distinct and on average less anorthitic in the Brandberg rocks (An_{20-30}) compared to the Sarusas/Khoraseb quartz latites (ca. An_{40-46}). This is consistent with the slightly higher degree of magmatic differentiation in the Brandberg granites, and there is no evidence that any of these plagioclase compositions are in disequilibrium with the bulk rock composition (Cornet *et al.*, 2022).
- (3) Brandberg granites are texturally isotropic, non-foliated rocks that show no textural evidence of crystal accumulation by settling or compaction.

Geochemical signatures of zircon crystals from the Brandberg main granites and the Sarusas/Khoraseb quartz latites show broad similarities, e.g. in their degrees of Eu anomalies (Fig. 10A) or Hf isotope compositions (Fig. 11). While the zircon crystals from the quartz latites have higher average Ti concentrations and Th/U ratios compared to those from the Brandberg main granites, these differences can stem from kinetic processes due to the fast growth rates of zircon in the volcanic environments (Kirkland *et al.*, 2015) or different magmatic temperatures (in the case of Ti; Watson *et al.*, 2006), and not from major differences from the compositions of the parent magmas.

These findings indicate that Brandberg granites are too evolved geochemically to represent the complementary cumulate left-overs from large volume quartz latites (here represented by the Sarusas/Khoraseb ignimbrites), and that the Brandberg magma reservoir did not experienced large-scale extraction of interstitial melts that fed volcanic eruptions. The Brandberg plutonic rocks and Northern Etendeka volcanics hence most likely represent melts of similar compositions that formed through similar differentiation processes, but ended up with a different fate by solidifying at different levels in the crust

(upper crust for the Brandberg, and Earth's surface for the quartz latites).

Possible origins of high-volume, hot, crystal-poor ignimbrites in the Etendeka province

The Paraná-Etendeka LIP produced some of the largest volume, hottest and most crystal-poor silicic magmas on Earth and highly unusual ignimbrite deposits that combine the crystal-poor character of extracted melt pockets in zoned ignimbrites (Hildreth & Wilson, 2007; Bachmann *et al.*, 2014; Forni *et al.*, 2016; Keller *et al.*, 2021) with the horizontal and vertical homogeneity typical for the crystal-rich monotonous intermediate ignimbrites, the latter representing rejuvenated bulk mush erupted from rapidly stirred magma chambers (Hildreth, 1981; Bachmann *et al.*, 2002; Maughan *et al.*, 2002; Huber *et al.*, 2012; Repstock *et al.*, 2022; Zhao *et al.*, 2023). Petrographically, they closely resemble the hotspot-related, unzoned, rheomorphic ignimbrites of the Snake River Plain (Ellis *et al.*, 2013).

One possibility to explain the unusual combination of characteristics is that some silicic reservoirs in the Northern Etendeka crust experienced extensive remelting/rejuvenation due to some unusually high flux of mafic recharge at a given point of the evolution of the system. Such a hypothesis would be supported by the bulk compositions of the Etendeka ignimbrites that are dacitic to rhyodacitic (62–70 wt % SiO₂; Ewart *et al.*, 1998b, 2004b), while high-silica rhyolites (>75 wt %) typical of melt extraction in the upper crust are absent. However, no clear resorption textures are observed in the phenocrysts (Fig. 5E), and the main phenocrysts present in these units (pyroxene, plagioclase, oxides) are typically refractory minerals that are difficult to remelt (Ellis *et al.*, 2023). Moreover, while geochemical modelling by Ewart *et al.* (2004b) shows that the late-erupted Sarusas quartz latite can be explained compositionally by hot mafic recharge mixing into the residual magma after the eruption of the Khoraseb quartz latite, the estimated volume of this mafic recharge (5–6%) is unlikely to be sufficient to cause large-scale mush remelting and remobilization.

A more likely possibility is that the Etendeka ignimbrites were extracted from magma reservoirs deeper than the current, upper-crustal exposure level in the Damaraland province (at least 3 km depth; Raab *et al.*, 2005). This could explain the observed lack of voluminous crystal cumulates expected in the Etendeka volcanic deposits: most of the fractionated crystals are likely trapped within cumulate deposits presently not exposed at the surface, apart perhaps from some gabbroic facies of the Messum pluton that show a positive Eu anomaly, suggesting crystal accumulation (Harris *et al.*, 1999). Application of precise geobarometric methods to the Etendeka pyroclastic deposits is hampered by the absence of an appropriate phenocryst phase assemblage, but haplogranite geobarometry (Blundy & Cashman, 2001) based on bulk-rock compositions of the crystal-poor ignimbrites (Fig. S17) indicates an origin from 200 to 800 MPa. A deep origin of the magma chamber is also supported by the lack of caldera structures during such large eruptions in the Etendeka province (Milner *et al.*, 1992); shallow accumulations of melt should always lead to a well-formed caldera, as observed in many areas around the world (Lipman, 1997). Furthermore, AFC modelling for the highly voluminous Sarusas/Khoraseb-type ignimbrites show that their magmatic compositions can be best explained by assimilation of middle-crust model compositions (Rudnick & Fountain, 1995; Taylor & McLennan, 1995) into primary mantle melts (Ewart *et al.*, 2004b). Considering the geochemical composition and mineralogy of the Northern Etendeka quartz latites, the plutonic roots left

behind by these eruptions should be (ferro-)gabbros, which show accumulation of clinopyroxenes, plagioclase, and Fe–Ti oxides. A possible analogue for such LIP-related, mafic plutonic systems, apart from the small gabbroic facies of the Messum pluton, is the Panzhihua gabbroic intrusion within the Permian Emeishan LIP (China), which shows clear evidence of crystal accumulations (in particular of Fe–Ti oxides; Pang *et al.*, 2008).

Estimation of magmatic volumes and fluxes

Magmatic flux is one of the key parameters that control whether magma batches are trapped as plutons in the crust or erupt to the surface. The growth and eruptibility of magma chambers are most effectively controlled by the rate of magma injection, and the coupled rheological response of the crust to the mechanical strain caused by the injected magma batches (Degruyter *et al.*, 2016; Huber *et al.*, 2019; Townsend *et al.*, 2019). Sustainable growth of magma chambers only occurs if fluxes are high enough to prevent a premature thermal death of the system, but low enough to prevent frequent draining of the chamber to the surface (Huber *et al.*, 2019).

To quantify the original volume of the Brandberg intrusion and the magmatic fluxes during its assembly, we assume a cylindrical shape for the magma body. Considering a surface area of ~650 km² based on satellite images, negligible erosional losses indicated by thermochronological models (Raab *et al.*, 2005; Margirier *et al.*, 2019), and geomagnetic models predicting a 3 km thick mass of granitoids (ca. 0.5 km granite, 2.5 km quartz monzonite) over the deep gabbroic roots that extend down to ~10 km depth (Vietor *et al.*, 2001), we obtain 1950 km³ as a minimum estimate for the volume of the silicic magmas of Brandberg. Thus, the whole magmatic volume of the Brandberg massif is at least in the same order of magnitude as the erupted volumes of the Northern Etendeka quartz latites (Bryan *et al.*, 2010, and references therein). Considering the ~1.4 Myr duration of incremental growth at Brandberg from our new age data, the silicic magmatic flux is estimated as 0.0014 km³/yr.

Radiogenic Sr and Nd isotope data from the Brandberg granites suggest a mantle origin for the magmas (Schmitt *et al.*, 2000), and the geomagnetic model of Vietor *et al.* (2001) requires substantial amounts of mafic rocks below the granitoids. Therefore, the total magma flux must consider the mafic roots as well as material loss through volcanic eruptions. The total thickness of the Brandberg complex including its unexposed mafic roots is ~12 km, which results in ca. 7800 km³ for the entire magmatic volume (silicic and mafic). No silicic volcanic deposits directly related to Brandberg have been found in the Damaraland region, so the volcanic fluxes cannot be directly constrained. Taking two extreme scenarios, either no eruptions at all, or complete evacuation of the magma chamber followed by complete refilling, we obtain a possible range for the total magma volume of 7800–15 600 km³ (silicic and mafic) and a total flux of 0.006–0.012 km³/yr. These are similar to lower than those typical of other caldera-related plutonic systems (see Table 6 of Lipman & Bachmann, 2015) and large caldera-forming volcanoes such as Laguna del Maule, Campi Flegrei, Santorini and Aso (Townsend *et al.*, 2019).

For the Northern Etendeka quartz latites, calculated fluxes are significantly larger compared the Brandberg. The Sarusas-Guarapuava-Tamarana and the Khoraseb-Ourinhos systems together produced minimum eruptive volumes of 8587 and 3929 km³ DRE, respectively (Bryan *et al.*, 2010, and references therein). The minimal durations of magmatic activity estimated from the zircon age range, including antecrysts, is 0.7 Myr for Sarusas and 0.9 Myr for Khoraseb, resulting in silicic magmatic

fluxes of ca. 0.0112 and 0.0044 km³/yr, respectively (Brandberg: 0.0014 km³/yr). The total magmatic fluxes for the Northern Etendeka quartz latites are difficult to constrain, as their plutonic roots are not exposed. Rhyolite models that combine AFC and crustal assimilation estimate the ratios between the volumes of silicic magmas produced and the volumes of parental mafic magmas required for their generation as ranging from 1:1 to 1:5 (Bryan *et al.*, 2008 and references therein). Using these values as minimum and maximum estimations for the size of the plutonic roots beneath the Northern Etendeka ignimbrites, we estimate total magmatic fluxes (silicic + mafic) for the Sarusas and the Khoraseb quartz latites as 0.022–0.067 km³/yr and 0.009–0.026 km³/yr, respectively. These total fluxes for the volcanic systems are, therefore, typically larger than the same fluxes calculated for the Brandberg pluton (0.006–0.012 km³/yr).

Effects of crust maturity on pluton formation in the Etendeka LIP

One of the main parameters for efficient growth of magma reservoirs is the thermal maturity of the crust, as a warmer crust leads to slower cooling, lower viscosity and thus a more accommodating environment for large magma reservoirs (de Silva *et al.*, 2006; Gregg *et al.*, 2012; Townsend *et al.*, 2019). In continental LIP-related magmatic systems, the large-scale emplacement and eruption of mafic melts during the main flood basalt activity leads to a thermally mature crust by the time silicic volcanism develops, which favors the build-up of large silicic magma chambers in the later stages of LIP evolution. This timing of mafic vs. silicic magmatism is, for instance, demonstrated in the Karoo LIP of southern Africa, where the rhyolites of the Lebombo volcanic margin clearly postdate the tholeiitic flood basalts based on ⁴⁰Ar–³⁹Ar ages and stratigraphic relationships (Miller & Harris, 2007). Similarly, the entrapment of large silicic plutons such as Brandberg or Erongo followed the main volcanic peak in the Paraná–Etendeka LIP (Schmitt *et al.*, 2000; Wigand *et al.*, 2004; Sun *et al.*, 2024, and references therein).

The volcanic to plutonic (V/P) ratio, given by the relative proportions of extrusive versus intrusive rocks emplaced at a given time, is an important parameter for the evolution of magmatic provinces (Bachmann & Huber, 2024). Comparison of the radiometric ages of the main volcanic and plutonic units in the Paraná–Etendeka LIP indicate a diminishing V/P during the evolution of the province. This can be explained by (1) a lower flux of magma recharge towards the end of magmatism and (2) progressive thermal maturation leading to a warmer and less viscous crust over time, which makes magma chamber over-pressurization much less efficient, and therefore prevents volcanic eruptions (Huber *et al.*, 2019). Inhibition of late volcanic activity by a decreasing magmatic flux is likely exacerbated in LIPs by the mineralogy of the dry tholeiitic melts, which contain more abundant plagioclase, Fe–Ti oxides and clinopyroxene. This requires a higher thermal input for mush remobilization from this typically refractory mineral assemblage, compared to the wetter granite melts prevalent in Cordilleran arcs, where phases with low melting temperatures such as K-feldspar and quartz are more abundant (Wolff *et al.*, 2015; Ellis *et al.*, 2023).

CONCLUSIONS

The Etendeka province in Namibia is a rare occurrence where coeval, high-volume magmatic systems representing two different crustal levels (silicic upper crustal reservoirs and erupted

ignimbrite deposits) can be studied within the context of a continental LIP. Our model for the Brandberg based on new CA-ID-TIMS zircon geochronology and field observations interprets it as a resurgent filling of a shallow reservoir that was emplaced in four different stages:

- (1) Localized subsidence formed a circular fault system around the pluton, interpreted as a caldera fault, indicating precursor, presumably eroded, volcanic activity prior to 133.2 Ma.
- (2) From 133.20 ± 0.19 to 132.62 ± 0.29 Ma (ca. 0.6 Myr), the main granite body was injected into preexisting fractures and ring faults related to the earlier caldera formation.
- (3) The intrusion of the main granite body was followed by a small-volume quartz monzonite plug (132.351 ± 0.092 to 132.14 ± 0.18 Ma) emplaced near the centre of the pluton, and by peripheral trachydacite dykes (132.733 ± 0.042 to 132.17 ± 0.51 Ma).
- (4) The magmatic activity ended with the final intrusion of a satellite peralkaline granite centre with associated pegmatites and aplite dykes from 131.910 ± 0.039 to 131.838 ± 0.043 Ma. This final stage was accompanied by hydrothermal activity caused by exsolved fluids from the crystallizing granitic magma ('auto-metasomatism'), leading to recrystallization and albitization of early-formed plagioclase and formation of granophyric quartz–K-feldspar aggregates in the Brandberg granitoids.

In the context of the LIP, plutonic entrapment of Brandberg followed the main mafic and earlier silicic phases of the Etendeka volcanism based on stratigraphy, and is largely coeval to the latest major phase of silicic eruptions occurring between 133.3 and 132.6 Ma with emplacement of the Khoraseb, Fria and Sarusas quartz latites. The assembly of Brandberg, however, ends later than the Khoraseb–Sarusas quartz latite volcanism and suggests that the entrapment of large, silicic plutons like Brandberg was part of the latest stage in the Paraná–Etendeka LIP evolution. The plutonic entrapment was aided by (1) waning magmatic fluxes from the mantle and (2) the establishment of a thermally mature upper crust heated by the voluminous high-temperature flood basalts that preceded silicic magmatism, leading to a more viscous and accommodating crustal environment for magma accumulation. The effect of waning fluxes on the eruption frequency is likely exacerbated by the refractory cumulate crystal assemblages (plagioclase, clinopyroxene, Fe–Ti oxides) found in the dry and hot, LIP or hotspot-related magmas, which require a higher thermal input to remelt and therefore require high recharge flux to erupt. Total magma fluxes (silicic + mafic) for the buildup of Brandberg are estimated as 0.006–0.012 km³/yr, generally similar to or lower than granitic batholiths from arc systems, and significantly lower compared to the volumetrically comparable Northern Etendeka silicic ignimbrites (Sarusas: 0.022–0.067 km³/yr; Khoraseb: 0.009–0.026 km³/yr).

These hot, homogeneous, and crystal-poor Northern Etendeka ignimbrites are not more evolved than the Brandberg granite geochemically, indicating that they were not extracted as interstitial melts from reservoirs resembling Brandberg in composition, but rather represent coeval analogue magmas formed by similar differentiation processes. The Northern Etendeka ignimbrites were likely extracted from mid-crustal magma reservoirs, producing cumulates more mafic in composition than the Brandberg pluton. Such cumulative mafic plutons are not exposed in the Northern Etendeka region at present, but may be represented by some gabbroic facies in the Messum Pluton (Southern Etendeka, which shows much deeper level of exposure), and can be found in other,

more deeply dissected LIPs in other places on Earth (e.g. Permian Emeishan LIP in China).

SUPPLEMENTARY DATA

Supplementary data are available at *Journal of Petrology* online.

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DATA AVAILABILITY

All original analytical data belonging to this manuscript are deposited in the EarthChem repository Sun, Y., Galli, A., Szymanowski, D., Guillong, M., Simon, J., Shipandeni, A. & Bachmann, O. (2025). Bulk rock major and trace element compositions, zircon U–Pb and Hf isotope compositions and zircon trace element compositions from granites and quartz latites from the Damaland/Etendeka region, Namibia, Version 1.0. Interdisciplinary Earth Data Alliance (IEDA), <https://doi.org/10.60520/IEDA/113413> (Accessed 12 May 2025) and are supplied in supplementary material 2.

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